PHYSICS

1) A bar magnet when placed at an angle of 30° to the direction of magnetic field induction of 5×10^{-2}T, experiences a moment of couple 25×10^{-6} N-m. If the length of the magnet is 5 cm, its pole strength is:-

(1) 2×10^{-2} A-m

(2) 5×10^{-2} A-m

(3) 2 A-m

(4) 5 A-m

2) Two identical short bar magnets, each having magnetic moment M, are placed at a distance 2d apart with their axes perpendicular to each other in a horizontal plane. The magnetic induction at a point midway between them is :-

(1) $\frac{\mu_0}{4\pi}(\sqrt{2})\frac{M}{d^3}$

(2) $\frac{\mu_0}{4\pi}(\sqrt{3})\frac{M}{d^3}$

(3) $\left(\frac{2\mu_0}{\pi}\right)\frac{M}{d^3}$

(4) $\frac{\mu_0}{4\pi}(\sqrt{5})\frac{M}{d^3}$

3) A wire of length L metre carrying a current of I ampere is bent in the form of a circle. Its magnitude of magnetic moment will be :

(1) $\frac{IL}{4\pi}$

(2) $\frac{IL^2}{4\pi}$

(3) $\frac{I^2L^2}{4\pi}$

(4) $\frac{I^2L}{4\pi}$

4) An electron is moving along positive x-axis. A uniform electric field exists towards negative y-axis. What should be the direction of magnetic field of suitable magnitude so that net force on electron is zero ?

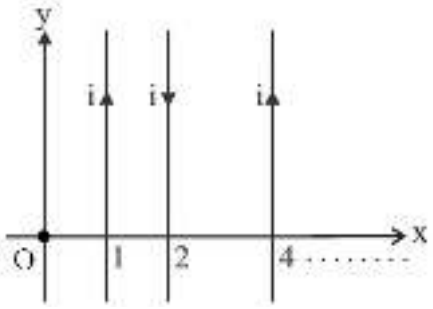
(1) positive z-axis

(2) negative z-axis

(3) positive y-axis

(4) negative y-axis

5) Equal currents, i = 1 A each, are flowing through the infinitely long wires placed parallel to y-axis located at x = + 1m, x = +2m, x = +4 m, ... etc. but in opposite directions as shown in figure. The magnetic field at origin (in tesla) would be :-



(1) $-1.33 \times 10^{-7}\,\hat{k}$

(2) $1.33 \times 10^{-7}\,\hat{k}$

(3) $2.67 \times 10^{-7}\,\hat{k}$

(4) $-2.67 \times 10^{-7}\,\hat{k}$

6) A thin flexible wire of length L is connected to two adjacent fixed points carries a current I in the clockwise direction, as shown in the figure. When system is put in a uniform magnetic field of strength B going perpendicularly into the plane of paper, the wire takes the shape of a circle. The tension in the wire is:-



(1) IBL

(2) $\frac{IBL}{\pi}$

(3) $\frac{IBL}{2\pi}$

(4) $\frac{IBL}{4\pi}$

7) A proton and α particle, after accelerating through same potential difference enters into a uniform magnetic field perpendicular to their velocities. Find the ratio of radius of path of proton and α-particle:-

(1) 1 : 2

(2) 1 : 1

(3) 2 : 1

(4) 1 : $\sqrt{2}$

8) A long cylindrical conductor of radius 'R' carries a current 'i'. The value of magnetic field at a point which is R/4 distance inside from the surface is 10 T. Find the value of magnetic field at a point

which is at 4R distance outside from the surface

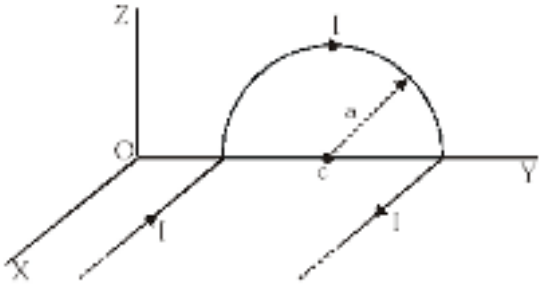
(1) $\frac{4}{3}T$

(2) $\frac{8}{3}T$

(3) $\frac{40}{3}T$

(4) $\frac{80}{3}T$

9) A long wire bent as shown in the figure carries current I. If the radius of the semicircular portion is a, the magnetic induction at the centre c is-



(1) $\frac{\mu_0 I}{4a}$

(2) $\frac{\mu_0 I}{4\pi a}\sqrt{\pi^2+4}$

(3) $\frac{\mu_0 I}{4a}+\frac{\mu_0 I}{2\pi a}$

(4) $\frac{\mu_0 I}{4\pi a}\sqrt{\pi^2-4}$

10) Considering magnetic field along the axis of a circular loop of radius 1 meter, at what distance 'x' from the centre of the loop the magnetic field is $\frac{1}{2\sqrt{2}}$ times of its value at the centre ?

(1) x = 1 m

(2) x = 2 m

(3) x = 3 m

(4) x = 4 m

11) Relative perimitivity and permeability of a material are ε_r and μ_r respectively. Which of the following values of these quantities are allowed for a diamagnetic material ?

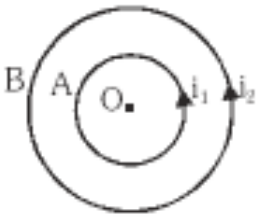
(1) $\varepsilon_r = 0.5, \mu_r = 1.5$

(2) $\varepsilon_r = 4.5, \mu_r = 0.5$

(3) $\varepsilon_r = 0.5, \mu_r = 0.5$

(4) $\varepsilon_r = 2.5, \mu_r = 1.5$

12) A and B are two concentric circular conductors of centre O and carrying current i_1 and i_2 as shown in the diagram. If ratio of their radii is 1 : 2 and ratio of the magnetic flux densities at O due



to A and B is 1:3, then the value of $\frac{i_1}{i_2}$ will be :–

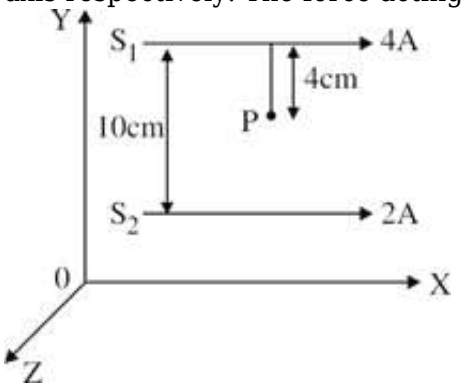
(1) $\frac{1}{2}$

(2) $\frac{1}{3}$

(3) $\frac{1}{4}$

(4) $\frac{1}{6}$

13) Two long parallel conductors S_1 and S_2 are separated by a distance 10 cm and carrying currents of 4A and 2A respectively. The conductors are placed parallel to x-axis in X-Y plane. There is a point P located between the conductors (as shown in figure). A charge particle of 3π coulomb is passing through the point P with velocity $\vec{v} = (2\hat{i} + 3\hat{j})\text{m/s};\ \text{where } \hat{i} \& \hat{j}$ represents unit vector along x & y axis respectively. The force acting on the charge particle is $4\pi \times 10^{-5}(-x\hat{i} + 2\hat{j})\text{N}$. The value of x is :



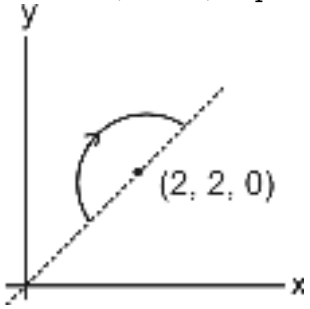
(1) 2

(2) 1

(3) 3

(4) –3

14) A uniform magnetic field $\vec{B} = 3\hat{i} + 4\hat{j} - \hat{k}$ exists in a region of space. A semi-circular wire of radius 2 m, carrying current $\sqrt{2}$ A, having its center at (2, 2, 0) is placed in x-y plane as shown in figure.



The force on semi-circular wire will be :

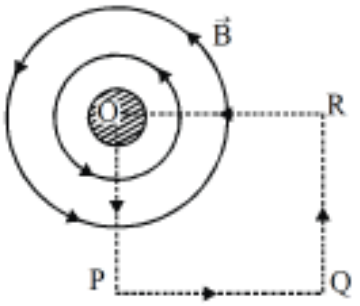
(1) $4\hat{i} + 4\hat{j} + 4\hat{k}$

(2) $-4\hat{i}+4\hat{j}+4\hat{k}$

(3) $4\hat{i}-4\hat{j}+4\hat{k}$

(4) $4\hat{i}+4\hat{j}-4\hat{k}$

15) A current carrying thick wire (current = i) perpendicular to the plane of the paper produces a magnetic field, as shown in the figure. A square of side *a* is drawn with one of its vertices on the centre of the wire. The integral $\int \vec{B}.d\vec{\ell}$ along OPQRO has the value



(1) $+\mu_0 i$

(2) $\frac{\mu_0 i}{8}$

(3) $\frac{\mu_0 i}{4}$

(4) $\frac{\mu_0 i}{2}$

16) Susceptibility of Mg at 300 K is 1.2×10^{-5}. The temperature at which susceptibility will be 1.8×10^{-5} is :

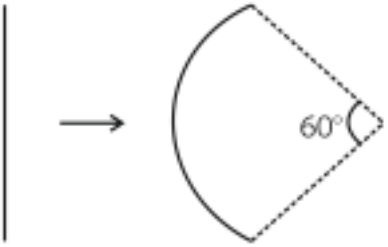
(1) 450 K

(2) 200 K

(3) 375 K

(4) None of these

17) A magnetised wire of moment M is bent into an arc of a circle subtending an angle of 60° at the centre; then the new magnetic moment is :-



(1) $(2M/\pi)$

(2) (M/π)

(3) $(3\sqrt{3}M/\pi)$

(4) $(3M/\pi)$

18) An electric bulb has a rated power of 50 W at 100 V. If it is used on an AC source of peak value 200 V, 50 Hz, a choke has to be used in series with it. This choke should have an inductance of : (Neglect resistance of choke coil)

(1) 0.1 mH

(2) 1 H

(3) 0.1 H

(4) 1.1 H

19) A bulb is connected first with dc and then ac of same voltage, it will shine brightly :-

(1) With AC

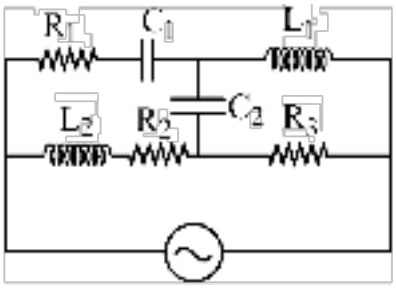
(2) With DC

(3) With having brightness in ratio 1/1.4

(4) Equally with both

20)

Find effective impedance at very high frequencies:-



(1) $R_1 + R_3$

(2) $\frac{R_1R_2}{R_1 + R_2} + R_3$

(3) $R_1 + R_2 + R_3$

(4) None of these

21)

The root mean square value of voltage, if an alternating voltage is given by $e = e_1 \sin \omega t + e_2 \cos \omega t$ is :-

(1) $\sqrt{\frac{e_1^2 + e_2^2}{2}}$

(2) $\frac{\sqrt{e_1^2 + e_2^2}}{2}$

(3) $\sqrt{\frac{e_1^2 - e_2^2}{2}}$

(4) $\frac{\sqrt{e_1^2 - e_2^2}}{2}$

22) What is the frequency of power in an A.C. circuit connected to a capacitor ?
(Given : f = 50 Hz, V = 220V)

(1) 50Hz

(2) 100Hz

(3) 25Hz

(4) 60Hz

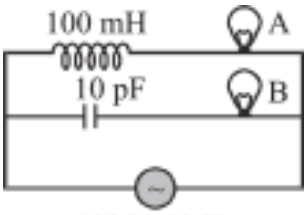
23) In an electrical circuit R, L, C and an AC voltage source all are connected in series. When L is removed from the circuit, the phase difference between the voltage and the current in the circuit is π/3. If instead, C is removed from the circuit, the phase difference is again π/3. The power factor of the circuit is

(1) 1/2

(2) $1/\sqrt{2}$

(3) 1

(4) $\sqrt{3}/2$



24) If A and B are identical bulbs, which bulb glows brighter ?

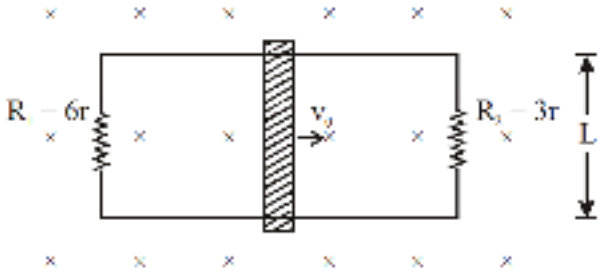
(1) A

(2) B

(3) Both equally bright

(4) Cannot say

25) A uniform rod of mass m is moving with constant velocity v_0 in a perpendicular uniform magnetic field B as shown. The resistance of rod is r. The current flowing through rod, R_1 and R_2 will be :-



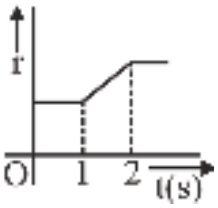
(1) $\frac{Bv_0L}{3r}, \frac{2}{3}\cdot\frac{Bv_0L}{3r}, \frac{1}{3}\cdot\frac{Bv_0L}{3r}$

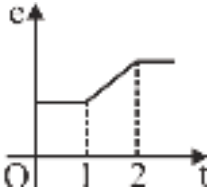
(2) $\frac{Bv_0L}{3r}, \frac{1}{3}\cdot\frac{Bv_0L}{3r}, \frac{2}{3}\cdot\frac{Bv_0L}{3r}$

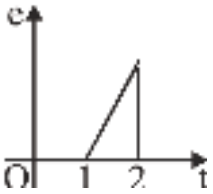
(3) $\frac{Bv_0L}{3r}, \frac{1}{3}\cdot\frac{Bv_0L}{3r}, \frac{1}{3}\cdot\frac{Bv_0L}{3r}$

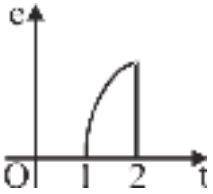
(4) $\frac{Bv_0L}{3r}, \frac{Bv_0L}{3r}, \frac{Bv_0L}{3r}$

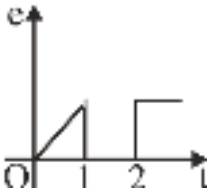
26) A circular coil is placed in uniform magnetic field such that its plane is perpendicular to field. The radius of coil changes with time as shown in the figure. Then which of the following graph

represent the induced emf in the coil with time- 

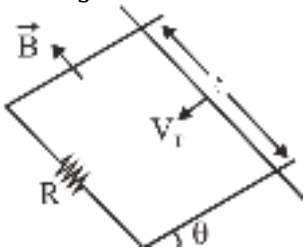
(1) 

(2) 

(3) 

(4) 

27) A copper rod of mass m slides under gravity on two smooth parallel rails distance apart set at an angle θ to the horizontal. At the bottom, the rails are joined by a resistance R.



There is a uniform magnetic field perpendicular to the plane of the rails. The terminal velocity of the rod is

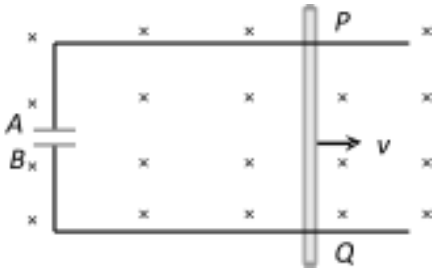
(1) $\frac{mgR\cos\theta}{B^2\ell^2}$

(2) $\frac{mgR\sin\theta}{B^2\ell^2}$

(3) $\frac{mgR\tan\theta}{B^2\ell^2}$

(4) $\frac{mgR\cot\theta}{B^2\ell^2}$

28) A conducting rod PQ of length L = 1.0 m is moving with a uniform speed v = 2 m/s in a uniform magnetic field (B = 4T) directed into the paper. A capacitor of capacity C = 10 μF is connected as



shown in figure. Then

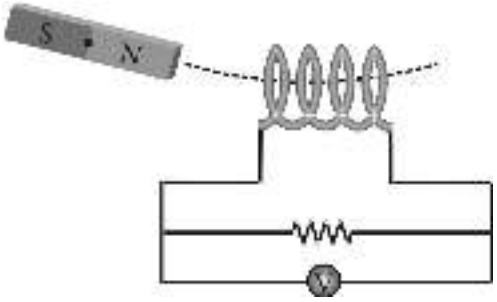
(1) $q_A = +80\ \mu C$ and $q_B = -80\ \mu C$

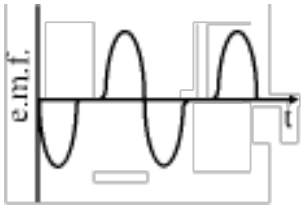
(2) $q_A = -80\ \mu C$ and $q_B = +80\ \mu C$

(3) $q_A = 0 = q_B$

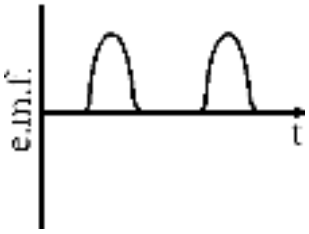
(4) Charge stored in the capacitor increases exponentially with time

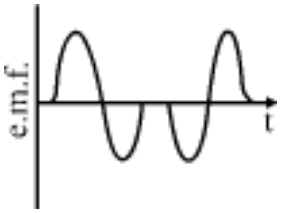
29) A magnet is made to oscillate with a particular frequency, passing through a coil as shown in figure. The time variation of the e.m.f. generated across the coil during one cycle is :-



(1) 

(2) 

(3) 

(4) 

30) A field line shown in the figure. This field line cannot represent - 

(1) Magnetic field
(2) Electrostatic field
(3) Induced electric field
(4) All of above

31) **Statement-I :** It is more difficult to push a magnet into a coil with more loops.
Statement-II : Emf induced in the coil resists the motion of the magnet.

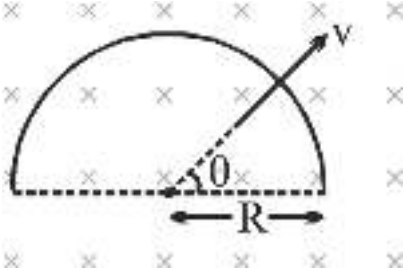
(1) Both Statement-I and Statement-II are true.
(2) Statement-I is false but Statement-II is true.
(3) Statement-I is true but Statement-II is false.
(4) Both Statement-I and Statement-II are false.

32)

Two conducting loops of radii R_1 and R_2 are placed in the same plane with their centres coinciding. If $R_1 >> R_2$, the mutual inductance M between them will be directly proportional to :

(1) $\frac{R_1}{R_2}$
(2) $\frac{R_2}{R_1}$
(3) $\frac{R_1^2}{R_2}$
(4) $\frac{R_2^2}{R_1}$

33) A semi circular loop of radius R is placed in a uniform magnetic field as shown. It is pulled with a



constant velocity. The induced emf in the loop is:

(1) Bv (πR) cosθ
(2) Bv(πR) sinθ
(3) Bv(2R) cosθ
(4) Bv(2R) sinθ

34) Some magnetic flux is changed from a coil of resistance 10Ω. As a result an induced current is developed in it which varies with time as shown in figure. The magnitude of change in flux through the coil (in weber) is :-



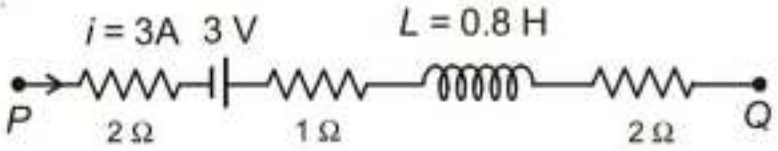
(1) 2

(2) 4

(3) 6

(4) None of these

35) Part of a circuit is shown. When current in circuit is 3A, rate of increase of current is 2 A/s. The potential difference between points P and Q at this instant is :



(1) 9.6V

(2) 10.6V

(3) 17.6V

(4) 13.6V

36) Match **List-I** with **List-II**. (The symbols carry their usual meaning.)

	List-I		**List-II**
A.	$\oint \vec{E}.d\vec{A} = \frac{Q}{\varepsilon_0}$	I.	Ampere-Maxwell's Law
B.	$\oint \vec{B}.d\vec{A} = 0$	II.	Faraday's Law
C.	$\oint \vec{E}.\vec{dl} = \frac{-d(\phi_B)}{dt}$	III.	Gauss Law of electrostatics
D.	$\oint \vec{B}.\vec{dl} = \mu_0 i_c + \mu_0 \varepsilon_0 \frac{d(\phi_E)}{dt}$	IV.	Gauss Law of magnetism

Choose the **correct** answer from the options given below :

(1) A-IV, B-III, C-II, D-I

(2) A-III, B-II, C-IV, D-I

(3) A-IV, B-I, C-III, D-II

(4) A-III, B-IV, C-II, D-I

37) A large parallel plate capacitor, whose plates have an area of 1 m^2 and are separated from each other by 1 mm, is being charged at a rate of 25 V s^{-1}. If the dielectric between the plates has the dielectric constant 10, then the displacement current at this instant is

(1) 25 μA

(2) 11 μA

(3) 2.2 μA

(4) 1.1 μA

38) For a plane electromagnetic wave propagating in the +Z direction, which one of the following combination gives the correct possible direction for $\vec{E}$ and $\vec{B}$ field respectively ?

(1) $\left(\hat{i}+2\hat{j}\right)$ and $\left(2\hat{i}-\hat{j}\right)$

(2) $\left(-2\hat{i}-3\hat{j}\right)$ and $\left(3\hat{i}-2\hat{j}\right)$

(3) $\left(2\hat{i}+3\hat{j}\right)$ and $\left(\hat{i}+2\hat{j}\right)$

(4) $\left(3\hat{i}+4\hat{j}\right)$ and $\left(4\hat{i}-3\hat{j}\right)$

39) An electromagnetic radiation has an energy 14.4 eV. To which region of electromagnetic spectrum does it belong?

(1) Ultraviolet region

(2) Visible region

(3) X-ray region

(4) γ-ray region

40) Which EM radiation is used in LASIK eye surgery

(1) Infra red

(2) Ultra violet

(3) Microwaves

(4) X-rays

41) In a wave amplitude of electric field E_0 = 100 Vm^{-1} then find the average value of Poynting's vector :-

(1) 13.25 Wm^{-2}

(2) 26.5 Wm^{-2}

(3) 18.25 Wm^{-2}

(4) 19.7 Wm^{-2}

42) In a circuit for finding the resistance of a galvanometer by half deflection method, a 6 volt battery and a high resistance of 11 kΩ are used. The figure of merit of the galvanometer is 60 μA/division. In the absence of shunt resistance, the galvanometer produces a deflection of θ = 9 divisions when current flows in the circuit. The value of shunt resistance that can cause deflection of θ/2 is closest to :

(1) 55 Ω

(2) 110 Ω

(3) 220 Ω

(4) 550 Ω

43) Current sensitivity of a moving coil galvanometer is 5 div/mA and its voltage sensitivity (angular deflection per unit voltage applied) is 20 div/V. The resistance of the galvanometer is :

(1) 40 Ω

(2) 25 Ω

(3) 250 Ω

(4) 500 Ω

44) Which of the following statement(s) is/are correct?
(A) A choke coil should have high inductance and low resistance
(B) A choke coil is preferred to a rehostat in A.C. circuits as it consumes almost zero power
(C) With increase in frequency of an AC supply, the inductive reactance decreases inversely with frequency
(D) With increase in frequency of an AC supply, the capacitive reactance increases directly proportional to frequency.

(1) ACD

(2) ABD

(3) ABCD

(4) AB

45) A charged particle enters a uniform magnetic field with velocity vector at an angle of 45° with the magnetic field. The pitch of the helical path followed by the particle is p. The radius of the helix will be :-

(1) $\frac{p}{\sqrt{2}\pi}$

(2) $\sqrt{2}p$

(3) $\frac{p}{2\pi}$

(4) $\frac{\sqrt{2}p}{\pi}$

CHEMISTRY

1) Match the column.

	Column-I (Ion)		**Column-II (Colour of flame)**
(A)	Ca^{+2}	(i)	Blood (crimson) red
(B)	Sr^{+2}	(ii)	Apple green
(C)	Ba^{+2}	(iii)	Brick red

Select the correct option :-

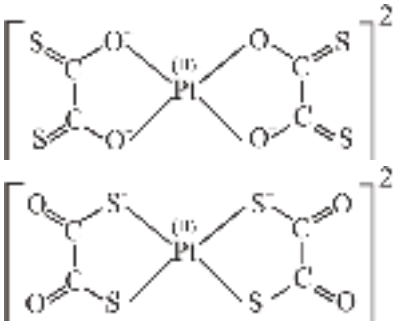
(1) A-ii, B-iii, C-i

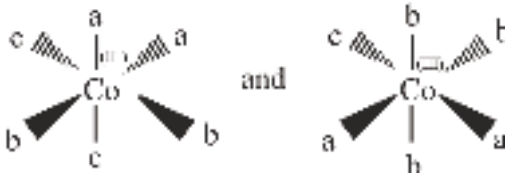
(2) A-iii, B-i, C-ii

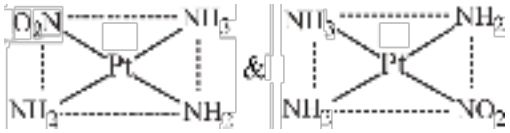
(3) A-iii, B-ii, C-i

(4) A-i, B-ii, C-iii

2) Identify the pair of complex which are stereoisomers of each other -

(1)  and

(2) 

(3) 

(4) All of the above

3) Which of the following statement is **not** correct?

(1) Copper liberates hydrogen from acids.

(2) In its higher oxidation states, Manganese forms compounds with oxygen and fluorine. e.g. MnO_3F

(3) Mn^{3+} and Co^{3+} are oxidising agents in aqueous solution.

(4) Ti^{2+} and Cr^{2+} are reducing agents in aqueous solution.

4) In a cyclic trimetaphosphoric acid molecule, how many double and single bonds are present ?

(1) 3 double bonds and 12 single bonds

(2) 6 double bonds and 12 single bonds

(3) 3 double bonds and 9 single bonds

(4) 6 double bonds and 9 single bonds

5) **Statement-I :** The chlorides of Be^{2+} and Mg^{2+} does not impart any colour to the flame test.
Statement-II : The hydration energy of Be^{2+} is higher than that of Mg^{2+}.

(1) Both **Statement I** and **Statement II** are incorrect.

(2) **Statement I** is correct but **Statement II** is incorrect.

(3) **Statement I** is incorrect but **Statement II** is correct

(4) Both **Statement I** and **Statement II** are correct.

6) Which of the following statement is **incorrect**.

(1) Oxidising power order → $SiCl_4 < SnCl_4 < PbCl_4$

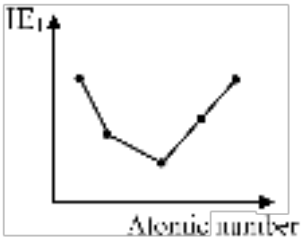
(2) Ionic character order → CsBr > RbBr > KBr > NaBr > LiBr

(3) The ionic character of lead (II) halides decreases with increase in atomic number of halogen

(4) The oxidation state of T□ in stable T□I_3 is +3

7) Select correct graph plotted between IE_1 of Group-13 and their atomic number :-

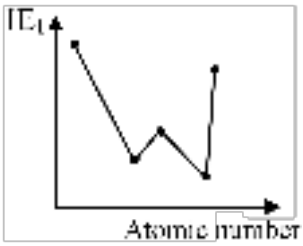
(1)



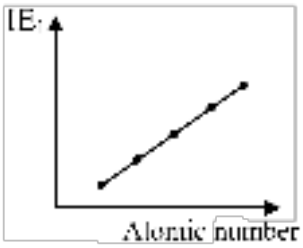
(2)



(3)



(4)



8) **Assertion (A)** : Correct order for the wavelengths of absorption in visible region for given complex is :
$[Ni(H_2O)_6]^{2+} > [Ni(NH_3)_6]^{2+} > [Ni(en)_3]^{2+}$
Reason (R) : According to spectrochemical series order of strength of ligand is $H_2O > NH_3 >$ en

(1) Both **(A)** and **(R)** are correct but **(R)** is not the correct explanation of **(A)**

(2) **(A)** is correct but **(R)** is not correct

(3) **(A)** is not correct but **(R)** is correct

(4) Both **(A)** and **(R)** are correct and **(R)** is the correct explanation of **(A)**

9) How many of the given compound form ppt
$BaSO_4$, $PbSO_4$, AgI, AgBr, AgCl, $SrSO_4$, $AgNO_3$, $CaCO_3$, $PbCrO_4$, $[Fe(H_2O)_5(NO)]$ SO_4

(1) 6

(2) 8

(3) 9

(4) 7

10) **Assertion (A) :** In octahedral complexes, electrons in the three orbitals (d_{xy},d_{yz},d_{zx}) are of low energy while the two orbitals ($d_{x^2-y^2}$,d_{z^2}) have high energy.
Reason (R) : In presence of asymmetrical field, The degeneracy of d-orbital of metal ion is maintained.

(1) Both Assertion and reason are correct and reason is correct for the Assertion.

(2) Both Assertion and Reason are correct but Reason is not correct for Assertion.

(3) Assertion is correct but Reason is incorrect.

(4) Assertion is incorrect but Reason is correct.

11) The major components in "Gun Metal" are:-

(1) Cu, Zn, Ni

(2) Cu, Sn, Zn

(3) Cu, Ni, Mg

(4) Al, Cu, Mg, Mn

12) Why HCl is not used to make the medium acidic in oxidation reactions of $KMnO_4$ in acidic medium?

(1) Both HCl and $KMnO_4$ act as oxidising agents.

(2) $KMnO_4$ oxidises HCl into Cl_2

(3) $KMnO_4$ is a weaker oxidising agent than HCl.

(4) $KMnO_4$ acts as a reducing agent in the presence of HCl.

13) Choose the incorrect statement.

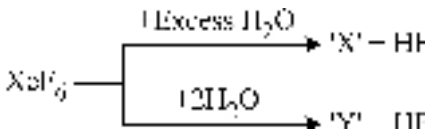
(1) The most common oxidation state of lanthanoids and actinoids is +3.

(2) In all lanthanoids and actinoids, the last electron is filled in ($\square - 2$) f subshell.

(3) Ions of lanthanoids with $4f^n$ configuration will have similar color to those having $4f^{14-n}$ configuration.

(4) In tripositive lanthanoid ions the number of unpaired electrons increase from Cerium to Gadolinium

14) Incorrect statement regarding following reactions is :



(1) 'X' is explosive

(2) 'Y' is an oxyacid of xenon

(3) Both are not example of redox reaction

(4) XeF_6 can undergo partial hydrolysis

15) In which of the following options all species contain X–O–X bonds in structure (X = central atom) ?

(1) $H_2S_2O_5$, S_3O_9, $S_2O_6^{-2}$

(2) P_4O_{10}, P_4O_6, $H_3P_3O_9$

(3) N_2O_5, N_2O, N_2O_4

(4) $H_4P_2O_7$, $H_4P_2O_6$, $H_4P_2O_5$

16) Which of the following is **not** true ?

(1) Among halide ions, iodide ion is the most powerful reducing agent

(2) Fluorine is the only halogen which does not show a variable oxidation state (in oxy acids)

(3) HOCl is stronger acid than HOBr

(4) HCl is stronger acid than HBr

17) Samples of which test tube when mixed gives white precipitate ?

(1) $BaCl_2$ + $AgNO_3$

(2) $BaCl_2$ + $ZnSO_4$

(3) $ZnSO_4$ + NaOH

(4) All of these

18) A complex has a composition corresponding to the formula $CoBr_2Cl.4NH_3$. What is the structural formula if conductance measurements show two ions per formula unit and Silver nitrate solution gives an immediate precipitate of AgCl but no AgBr :-

(1) $[CoBrCl(NH_3)_4]Br$

(2) $[CoCl(NH_3)_4]Br_2$

(3) $[CoBr_2Cl(NH_3)_4]$

(4) $[CoBr_2(NH_3)_4]Cl$

19) Which of the following facts about the complex $[Cr(NH_3)_6]Cl_3$ is **wrong**?

(1) The complex involves d^2sp^3 hybridisation and is octahedral in shape.

(2) The complex is paramagnetic.

(3) The complex is an outer orbital complex.

(4) The complex gives white precipitate with $AgNO_3$ solution.

20) A solution of white crystals gives a precipitate with $AgNO_3$ but no precipitate with a solution of Na_2CO_3. The action of conc. H_2SO_4 on the crystals yields a brown gas. The crystals is :-

(1) $NaNO_3$

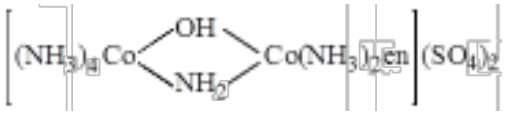
(2) KCl

(3) $Ca(NO_3)_2$

(4) NaBr

21)

Oxidation state and coordination number of Cobalt in given complex –

 are -

(1) +3 & 4

(2) +3 & 6

(3) +2 & 4

(4) +2 & 6

22) In context of the lanthanoids, which of the following statement is **not** correct ?

(1) Because of almost similar size or properties the separation of lanthanoids is not easy

(2) Availability of 4f electrons results in the formation of compounds in +4 state for all the members of the series

(3) Generally there is a gradual decrease in the radii of the members with increasing atomic number in the series

(4) All the members exhibit +3 oxidation state

23)

Match the column

	Compound		**No. of σ & π Bonds**
(a)	$H_2S_2O_3$	(P)	6 σ & 2 π
(b)	H_2SO_5	(Q)	11 σ & 4 π
(c)	$H_2S_2O_8$	(R)	9 σ & 4 π
(d)	$H_2S_2O_6$	(S)	7 σ & 2 π

(1) (a) S (b) P (c) Q (d) R

(2) (a) P (b) S (c) Q (d) R

(3) (a) P (b) Q (c) R (d) S

(4) (a) Q (b) S (c) P (d) R

24) BCl_3 does not exist as dimer but BH_3 exists as dimer (B_2H_6), because :-

(1) Chlorine is more electronegative than hydrogen

(2) There is pπ-pπ back bonding in $BC\square_3$ but BH_3 does not contain such bonding

(3) Large sized chlorine atoms do not fit between the small boron atoms whereas small sized hydrogen atoms get fitted between boron atoms.

(4) None of these

25) Which of the following statement is **wrong** :-

(1) Single N–N bond is stronger than the single P–P bond.
(2) PH_3 can act as a ligand in the formation of coordination compound with transition elements
(3) NO_2 is paramagnetic in nature
(4) Covalency of nitrogen in N_2O_5 is four

26) When I_2 vapours are passed through solutions of KCl, KF and KBr respectively, then :

(1) Cl_2 and Br_2 are evolved
(2) Cl_2 is evolved
(3) Cl_2, Br_2 and F_2 are evolved
(4) none of the above

27) Choose the correct statement from following ?

(1) He is a non-inflammable and heavy gas.
(2) Ar is used mainly to provide inert atmosphere in electric bulbs.
(3) XeO_3 is colourful explosive solid.
(4) Ne has maximum boiling point among inert gases.

28) Which of the following statement is correct ?

(1) Mc is a synthetic non-radioactive element.
(2) $NaNO_3$ is Indian saltpetre and KNO_3 is Chile saltpetre
(3) Density order ⇒ N > P > As > Sb > Bi
(4) E–H distance ⇒ $NH_3 < PH_3 < AsH_3$

29) Find correct match :

	Column-I (Compound)		**Column-II (Released gas on Hydrolysis)**
(a)	CaC_2	(i)	CH_4
(b)	Li_3N	(ii)	NH_3
(c)	$B_3N_3H_6$	(iii)	C_2H_2
(d)	Al_4C_3	(iv)	H_2

(1) a-iii, b-ii, c-iv, d-i,ii
(2) a-i, b-ii, c-iv, d-iii
(3) a-iii, b-ii, c-ii, iv, d-i
(4) a-i, b-ii, c-ii, iv, d-iii

30)

	Column-I		Column-II
(p)	$ZnCl_2$ + H_2S (basic)	(a)	Blood red colour
(q)	$CuSO_4$ + NH_4OH (excess)	(b)	White ppt form.
(r)	$FeCl_3$ + KSCN sol^n	(c)	deep blue solution
(s)	$Pb(NO_3)_2$ + KI	(d)	Yellow ppt form

(1) p-c, q-a, r-d, s-b

(2) p-b, q-c, r-a, s-d

(3) p-d, q-b, r-a, s-c

(4) p-b, q-a, r-c, s-d

31) What would you observe if you add excess dil. NaOH solution to $ZnCl_2$ solution ?

(1) A white ppt.

(2) A white ppt. which later dissolves

(3) A green ppt.

(4) A green ppt. which later dissolves

32) ***Assertion :*** Li_2SO_4 do not form double salt like alum.
Reason : Atomic size of Li is too small cannot accommodate 6 H_2O molecule.

(1) Both **(A)** and **(R)** are correct but **(R)** is not the correct explanation of **(A)**

(2) **(A)** is correct but **(R)** is not correct

(3) **(A)** is incorrect but **(R)** is correct

(4) Both **(A)** and **(R)** are correct and **(R)** is the correct explanation of **(A)**

33) **Assertion (A) :-** NO_2 being odd electron molecule dimerises at high temperature.
Reason (R) :- On dimerisation, NO_2 is converted to stable N_2O_4 molecule.

(1) Both (A) and (R) are true and (R) is the correct explanation of (A)

(2) Both (A) and (R) are true and (R) is not the correct explanation of (A)

(3) (A) is true but (R) is false

(4) (A) is false but (R) is true

34) Find the correct match :-

	Compound	Property
(1)	SO_3	Turns acidified $K_2Cr_2O_7$ green.
(2)	H_2S	Can act as oxidising agent
(3)	$HClO_2$	This oxy acid is stable in aqueous form

(4)	Cl_2	Disproportionates fastly in strongly acidic medium.

(1) 1

(2) 2

(3) 3

(4) 4

35) **Assertion (A) :** Copper (I) compounds are unstable in aqueous solution and undergo disproportionation.
Reason (R) : Cu^{2+}(aq.) is more stable than Cu^{+}(aq.) due to much more negative $\Delta_{hyd}H°$ of Cu^{2+}(aq.) which more than compensates for second ionisation enthalpy of Cu.
Which of the following option is correct-

(1) (A) is correct but (R) is incorrect.

(2) (A) is incorrect but (R) is correct.

(3) Both (A) and (R) are correct and (R) is the correct explanation of (A).

(4) Both (A) and (R) are correct but (R) is not the correct explanation of (A).

36) Which of the following is correct option ?

(1) $BF_3 < BCl_3$ (Lewis Acidic strength)

(2) $OF_2 < OCl_2$ (Bond angle)

(3) $NH_2(SiH_3) < NH(SiH_3)_2$ (N–Si Bond length)

(4) All of the above

37) In acidic solution of sodium nitroprusside when H_2S gas is passed, then -

(1) Violet colour form

(2) Violet ppt form

(3) H_2S will form 'S'

(4) No colour observed

38) Why should we avoid prolonged heating while preparing crystals of ferrrous ammonium sulphate ?

(1) to increase solubility of compound

(2) to prevent oxidation of ferrous ions

(3) to remove impurities from compound

(4) to speed up the crystallization process

39) The correct order of calculated spin-only magnetic moments of complex (A) to (D) is :-
(A) $[Ni(CO)_4]$ (B) $[Ni(H_2O)_6]Cl_2$
(C) $Na_2[Ni(CN)_4]$ (D) $[PdCl_2(PPh_3)_2]$

(1) (A) $\simeq$ (C) < (B) $\simeq$ (D)

(2) (C) $\simeq$ (D) < (B) $\simeq$ (A)

(3) (C) < (D) < (B) < (A)

(4) (A) ≃ (C) ≃ (D) < (B)

40) Which of the following ions does not liberate hydrogen gas on reaction with dilute acids ?

(1) Ti^{+2}

(2) Mn^{+2}

(3) Cr^{+2}

(4) V^{+2}

41) Which of the following is correct order of stability ?

(1) $Ga^{+} < Ga^{+3}$

(2) $Bi^{+3} < Sb^{+3}$

(3) $Sn^{+2} > Sn^{+4}$

(4) $Pb^{+2} < Pb^{+4}$

42) $E^{\ominus}/V$ for (M^{+3}/M) is positive for which element of group 13 ?

(1) Al

(2) Ga

(3) In

(4) Tl

43) Aqueous solution of sodium chromate when treated with conc. HNO_3 :-

(1) It is oxidised to Cr^{+7}

(2) It is reduced to Cr^{+3}

(3) $Cr_2O_7^{-2}$ and H_2O are produced

(4) $Cr_2O_7^{-2}$ and Cr^{+3} are produced

44) **Statement-1 :-** Test of NH_4^{+} can not be done with in group analysis.
Statement-2 :- During group analysis several times NH_4^{+} compound is added at the different steps.

(1) Both **Statement 1** and **Statement 2** are incorrect.

(2) **Statement 1** is correct but **Statement 2** is incorrect.

(3) **Statement 1** is incorrect but **Statement 2** is correct.

(4) Both **Statement 1** and **Statement 2** are correct.

45) Sum of total possible isomers of $[Pt(en)(NO_2)_2]$ and $[Pd(NH_3)_2(SCN)_2]$ are :-

(1) 8

(2) 9

(3) 10

(4) None

BIOLOGY

1) In the cross AaBb × Aabb, the genotype AABb: AaBB: aabb and aaBb will be obtained in the ratio :-

(1) 1 : 0 : 2 : 1
(2) 2 : 1 : 4 : 0
(3) 1 : 0 : 1 : 1
(4) 1 : 2 : 1 : 2

2) A test cross enables us to -

(1) determine the viability of cross
(2) distinguish between homozygous dominant and heterozygous dominant
(3) determine whether two species can interbreed
(4) determine the similarities in the DNA of two species

3) In pea plant, genotype (Bb) shows -

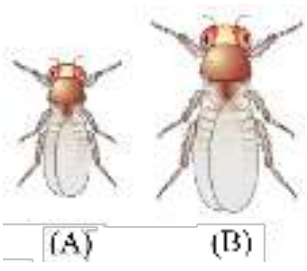
(1) Round seed with large sized starch grain
(2) Round seed with intermediate sized starch grain
(3) Round seed with small sized starch grain
(4) Wrinkled seed with small sized starch grain

4) A non-inheritable environmentally induced modification resembling the phenotype of a known gene, refers to :-

(1) Phenocopy
(2) Genocopy
(3) Genotype
(4) Phenotype

5) Which is not true for Mendal's law of dominance?

(1) Characters are controlled by discrete units called factors
(2) In dissimilar pair of factor one allele dominates over the other recessive allele.
(3) Factors occur in pairs
(4) Factors reach successive generations in their parental combination



6) How many statements are correct for given figure. (a) It is male and female

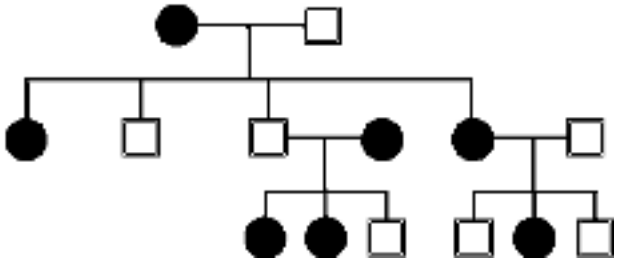
Drosophila melanogaster respectively.
(b) A small fruit fly.
(c) Complete their life cycle in about two weeks.
(d) Show sexual dimorphism
(e) With many types of hereditary variations
(f) It contains 4 pairs of chromosomes (2n = 8)
(g) Male with XY and female with XX

(1) 4
(2) 5
(3) 6
(4) 7

7) If recombination frequency between genes a and c is 2%, b and c is 13%, b and d is 4%, a and b is 15%, c and d is 17% and a and d is 19%. The sequence of genes in a chromosome is:

(1) acbd
(2) abcd
(3) dbac
(4) adbc

8) Given pedigree show inheritance of autosomal dominant disorder (Myotonic dystrophy). Then what will be genotype of shaded symbol :



(1) AA
(2) Aa
(3) AA or Aa
(4) aa

9) Identify the correct match from the column I,II and III :-

Column-I	Column-II	Column-III
(1) Monohybrid cross	(a) Tt×tt	(i) 9:3:3:1
(2) Dihybrid cross	(b) Tt×TT	(ii) 3:1
(3) Test cross	(c) TT×tt	(iii) 1:1
(4) Back cross	(d) RRYY × rryy	(iv) All tall

(1) 1-a-ii, 2-d-i, 3-c-iv, 4-b-iii
(2) 1-b-iii, 2-a-i, 3-c-ii, 4-d-iv
(3) 1-c-iv, 2-d-ii, 3-b-iii, 4-a-i

(4) 1-c- ii, 2- d- i, 3-a-iii, 4-b-iv

10) Which of the following trait is only express in homozygous condition -
(a) Yellow seed colour
(b) Green pod colour
(c) Inflated pod shape
(d) Terminal flower position
(e) Yellow pod colour

(1) Only e
(2) a, b, c
(3) a, b, c, d, e
(4) d and e

11) A man of A blood group marries a woman of AB blood group. which type of progeny would indicate that man is heterozygous A :-

(1) AB
(2) A
(3) O
(4) B

12) A normal woman, whose father was colour–blind is married to a normal man. The sons would be

(1) All colour–blind
(2) 75% colour–blind
(3) 50% colour–blind
(4) All normal

13) In cross between the pea plants homozygous for Round and Yellow seeds and Wrinkled and Green seeds, the ratio between the plants having Round seeds and Wrinkled seeds in F_2 generation will be -

(1) 9 : 7
(2) 3 : 1
(3) 1 : 1
(4) 2 : 1

14) Which of the following statements are correct ?
(a) A cross between F_1 hybrid and a recessive parent gives the ratio of 1 : 1.
(b) If the heterozygous tall pea plant is crossed with homozygous dwarf plant the proportion of tall progeny will be 50%.
(c) tt × tt cross will give recessive progeny in F_1 generation.
(d) In a cross between a male and female, both heterozygous for sickle cell anaemia gene, then percentage of diseased progeny will 75%.

(1) a, b, c, d

(2) a, b, d

(3) a, b, c

(4) b, c, d

15) Which one of the following statement is incorrect ?

(1) Physical association of genes on a chromosome is known as linkage.

(2) In birds females have one Z and one W chromosome besides the autosomes.

(3) Grasshopper males have two X-chromosome besides the autosomes.

(4) Alfred Sturtevant used the frequency of recombination between genes pairs as a measure of distance between genes.

16) As per ABO blood grouping system, the blood group of father is B, mother is A and child is AB. Their respective genotype can be :-
A. I^Bi / I^Ai / i i
B. I^BI^B / I^AI^A / i i
C. I^AI^B / i I^A / I^Bi
D. I^Ai / I^Bi / I^Ai
E. I^Bi / i I^A / I^AI^B
Choose the most appropriate answer from the options given below :-

(1) A only

(2) B only

(3) C and B only

(4) E only

17) **Assertion (A) :-** Based upon the observation on dihybrid crosses Mendel proposed law of independent assortment which states that when two pair of traits are combined in a hybrid, segregation of one pair of traits is independent of the other pair of traits.
Reason (R) :- In Mendel's dihybrid crosses each pair of trait segregated in a 3 : 1 ratio just like in a monohybrid cross.

(1) Both A and R are correct and R is correct explanation of A.

(2) Both A and R are correct but R is not correct explanation of A.

(3) A is correct but R is incorrect.

(4) Both A and R are incorrect.

18) **Assertion :** DNA cannot serve directly as a template for protein synthesis.
Reason : m-RNA carries the genetic information from nucleus to cytoplasm.

(1) Both Assertion & Reason are True & the Reason is a correct explanation of the Assertion.

(2) Both Assertion & Reason are True but Reason is not a correct explanation of the Assertion.

(3) Assertion is True but the Reason is False.

(4) Both Assertion & Reason are False.

19) **Statement-I :** DNA acts as the genetic material in all organisms.

Statement-II : RNA mostly functions as a messenger.

(1) Statement-I & Statement-II both are correct.
(2) Statement-I is correct & Statement-II is incorrect.
(3) Statement-I is incorrect & Statement-II is correct.
(4) Both Statement-I & Statement-II is incorrect.

20) If the sequence of bases in the coding strand of a doube stranded DNA is 5' GTTC GAGTC-3' the sequence bases in its mRNA will be.

(1) 5'-GACUCGAAC-3'
(2) 5'-CAAGCUCAG-3'
(3) 5-GUUCGAGUC-3'
(4) 3'GUUCGAGUC-5'

21) **Statement-I:-** In prokaryotes, such as *E.coli* they do not have a defined nucleus so DNA is scattered throughout the cell.
Statement-II:- In prokaryotes, there is a set of positively charged proteins called histones.

(1) Statement I and II both are correct
(2) Statement I and II both are incorrect
(3) Only Statement I is correct
(4) Only Statement II is correct

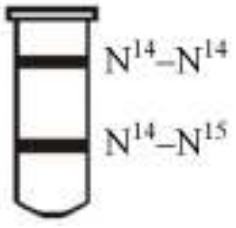
22) Adenosine is :

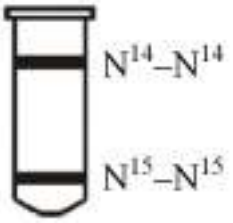
(1) A nitrogenous base and phosphate
(2) A nucleotide
(3) A ribonucleoside
(4) A ribonucleotide

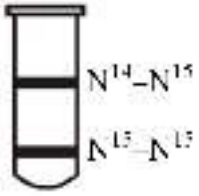
23) The structure of DNA is characterized by :

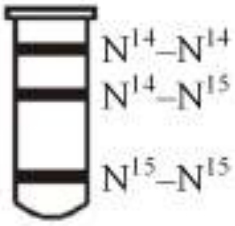
(1) Right-handed, double helix and antiparallel strands
(2) Right-handed, triple helix
(3) Right-handed, single helix and parallel strands
(4) Left-handed, double helix and parallel strands

24) Light DNA molecule is replicated in a heavy medium. After the four successive replication of DNA, 16 DNA molecules are formed. Which one the following is true for these DNA?

(1) 

(2) 

(3) 

(4) 

25) Which of the following represent the G-C rich DNA?

(1) Base ratio = 1.53

(2) Base ratio = 1.63

(3) Base ratio = 2.53

(4) Base ratio = 0.53

26) The salient feature of the DNA are :
(i) It is made of two polynucleotide chain
(ii) Backbone is constituted by sugar & nitrogen base
(iii) Two chains have parallel polarity
(iv) Bases in two strands are paired through hydrogen bond

(1) (i), (iv)

(2) (i), (ii), (iii)

(3) (i), (ii), (iv)

(4) (ii), (iv)

27) Under which of the following conditions will there be **no** change in the reading frame of following mRNA ?
5' UAC AGC GGU GCU AUA 3'

(1) Insertion of A at 5th position

(2) Insertion of A and G at 4th and 5th position respectively.

(3) Deletion of GGU from 7th, 8th and 9th positions respectively.

(4) Deletion of G from 5th position

28) Which of the following does not represent the correct result obtained by Griffith in his experiments ?

(1) S-strain → Inject into mice → Mice die

(2) R-strain → Inject into mice → Mice die

(3) S-strain (heat killed) → Inject into mice → Mice live

(4) S-strain (heat killed) + R-strain (live) → Inject into mice → Mice die

29) Which of the following is a wrong statement ?

(1) Histone proteins are rich in basic amino acids.

(2) Positively charged DNA is wrapped around negatively charged histone octamer.

(3) The nucleosomes in a chromatin are seen as 'beads-on-string' structure when viewed under electron microscope (EM).

(4) In DNA, the distance between two consecutive base pairs in 0.34 nm.

30) An *E.coli* with heavy DNA molecule is transferred into the medium with normal nitrogen ($^{14}NH_4C$□). After 40 minutes what will be the percentage of heavy DNA molecule?

(1) 12.5%

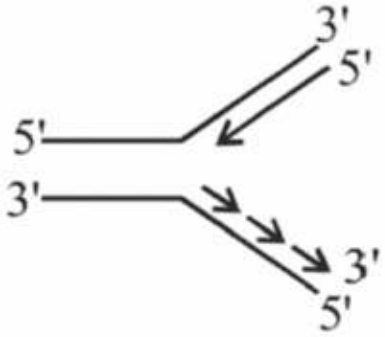
(2) 25%

(3) 0%

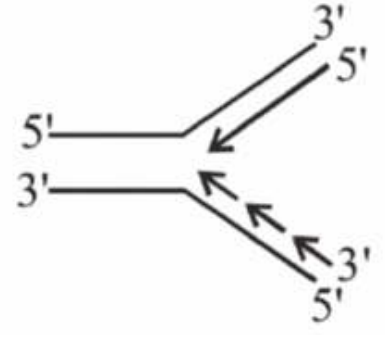
(4) 6.25%

31) Which of the following correctly represent DNA replication ?

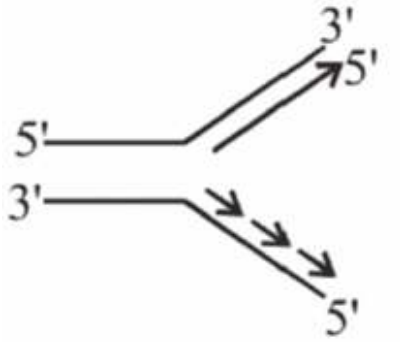
(1)



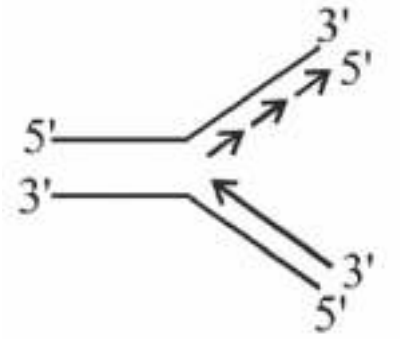
(2)



(3)



(4)



32) **Statement I :** Insertion or deletion of one or two bases changes the reading frame from the point of insertion but not from deletion.
Statement II : Insertion or deletion of three or its multiple bases insert or delete in one or multiple codon hence one or multiple amino acids and reading frame remains unaltered from that point onwards.

(1) Both **Statement I** and **Statement II** are incorrect.

(2) **Statement I** is correct but **Statement II** is incorrect.

(3) **Statement I** is incorrect but **Statement II** is correct.

(4) Both **Statement I** and **Statement II** are correct.

33) Main steps in DNA fingerprinting are given below. Arrange these steps in a correct sequence :-
A. Separation of DNA fragments by electrophoresis.
B. Blotting of separated DNA fragments to nylon membrane
C. Digestion of DNA by restriction
D. Hybridisation using labelled VNTR probe
Choose the correct answer from the options given below :-

(1) A, D, C, B

(2) D, A, B, C

(3) C, A, B, D

(4) C, A, D, B

34) Correctly match column-I, II and III :

	Column-I		**Column-II**		**Column-III**
A	Taylor	P	RNA polymerase III	i	Permease

B	Eukaryotes	Q	*Vicia faba*	ii	tRNA
C	AUG	R	Initiator codon	iii	Radioactive thymidine
D	Lac operon	S	Lac-y-gene	iv	Methionine

(1) A-P-i, B-Q-ii, C-R-ii, D-S-iv

(2) A-Q-ii, B-P-iii, C-R-iv, D-S-i

(3) A-Q-iv, B-R-ii, C-P-iii, D-S-i

(4) A-Q-iii, B-P-ii, C-R-iv, D-S-i

35) In *E.coli* strain, i gene gets mutated and its product cannot bind to the operator region in lac operon.
Select **correct** set of **statements**.
(a) Inducer cannot bind to i gene product.
(b) Small amount of lactose can enter cell.
(c) In absence of Glucose, z, y, a genes will be transcribed.
(d) i gene product will not affect binding of RNA polymerase to promoter.

(1) All Except (d)

(2) All Except (a)

(3) a, b only

(4) c, d only

36) Consider the following statements :
Statement-I : Severo ochoa enzyme helpful in polymerising DNA in a template independent manner.
Statement-II : In prokaryotes, control of the rate of transcriptional initiation is the predominant site for control of gene expression.

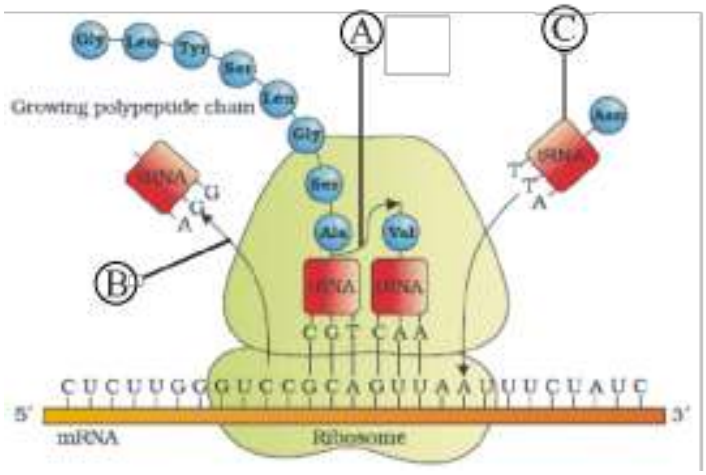
(1) Both statement-I and Statement-II are incorrect

(2) Statement-I is correct while statement-II is incorrect

(3) Both statement-I and statement-II are correct

(4) Statement-I is incorrect while statement-II is correct

37) The figure below shows three steps (A, B and C) of translation. Identify what they represent.

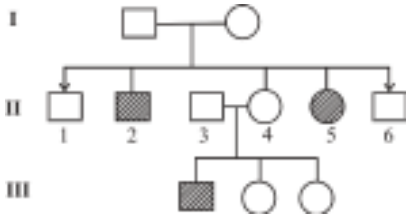


(1) A represents removal of t-RNA from E site and B represents peptide bond formation.

(2) B represents removal of t-RNA from E site and A represents formation of peptide bond.

(3) C represents arrival of I^{st} amino acid containing t-RNA at A site.

(4) A, B and C show termination reaction.

38) 

What is the genotype of II (3) :-

(1) AA

(2) aa

(3) X^{CY}

(4) Aa

39) Read the following statements and identify correct ones :-
(I) Gene that controls seed shape in garden pea is a pleiotropic gene.
(II) Starch grain size in garden pea seeds is a codominant character.
(III) Gene that control starch grain size shows monogenic quantitative inheritance.
(IV) Size of starch grain in garden pea seed is regulated by SBE gene.

(1) I, II and III

(2) I, III and IV

(3) Only II

(4) I, II, III and IV

40) Heterozygous individual for sickle-cell anaemia have how much percent of probability of transmission of the mutant gene to the progeny?

(1) 25%

(2) 50%

(3) 75%

(4) 100%

41) Mendelian disorder is :-

(1) Sickle cell anaemia

(2) Down syndromme

(3) Edward syndrome

(4) Turner syndrome

42) If the base sequence of a codon in mRNA is 5'-AUG-3', the sequence of anticodon of tRNA pairing with it must be :-

(1) 5'-AUG-3'

(2) 5'-UAC-3'

(3) 5'-CAU-3'

(4) 5'-GAU-3'

43) RNA act as genetic material in :-

(1) ϕ × 174 Bacteriophage

(2) λ bacteriophage

(3) *E. coli*

(4) QB bacteriophage

44)

Read the following statements and select the correct option :-

(1) Discontinous DNA strand is synthesized on DNA template strand with polarity 3' → 5'

(2) Euchromatin is loosely packed and stains dark.

(3) Process of transformation is not affected by RNase, protease and lipase enzyme but is affected by DNase enzyme.

(4) Hexose sugar is found in nucleic acid

45) Which of the following is nucleoside of RNA ?

(1) Cytosine

(2) Guanosine

(3) Adenylic acid

(4) Deoxycytidine

46) In EcoRI (restriction endonuclease) the letter 'R' is derived from :

(1) Genus of bacteria

(2) Species of bacteria

(3) Strain of bacteria

(4) Class of bacteria

47)

During isolation, precipitated DNA can be separated out by a process called _____.

(1) Elution

(2) Spooling

(3) Electrophoresis

(4) Biolistics

48) A foreign DNA is ligated at BamHI site in P^{BR322} vector. The recombinant plasmid will lose ..... resistance due to insertion of foreign DNA.

(1) Ampicillin

(2) Kanamycin

(3) Tetracycline

(4) Chloramphenical

49) The colonies of recombinant bacteria appear white in contrast to blue colonies of non-recombinant bacteria because of:

(1) Inactivation of glycosidase enzyme in recombinant bacteria.

(2) Non-recombinant bacteria containing beta-galactosidase.

(3) Insertional inactivation of alpha-galactosidase in non-recombinant bacteria.

(4) Insertional inactivation of beta-galactosidase in recombinant bacteria.

50) Which of the following is not a property of good cloning vector DNA :-

(1) Origin of replication (ori)

(2) It contain a selectable marker

(3) Single recognition site for commonly used restriction enzyme

(4) Many recognition sites for commonly used restriction enzyme

51) Which compound is used to stain DNA in Agarose Gel ?

(1) Ethidium bromide

(2) $NaNO_3$

(3) Ammonium nitrate

(4) Polyethylene glycol

52) **Assertion (A) :** *Agrobacterium tumifaciens* infects all broad leaved agriculture crops such as tomato, soyabean, sunflower, cotton and cereals.
Reason (R) : The Ti-plasmid of *agrobacterium tumifaciens* is responsible for inducing crown gall disease in plants.

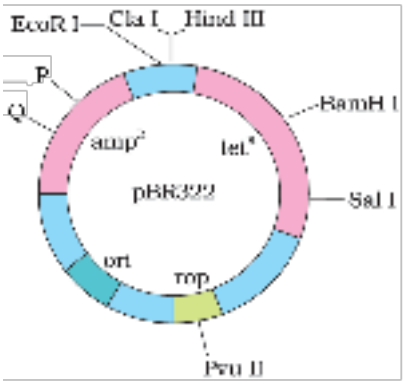
(1) Both (A) and (R) is correct and (R) is the correct explanation of (A).

(2) Both (A) and (R) is correct but (R) is not the correct explanation of (A).

(3) (A) is correct but (R) is incorrect.

(4) (A) is incorrect but (R) is correct.

53) P and Q in P^{BR322}, shown in the diagram given below, respectively represent recognition



sequences of -

(1) BamH I and Sal I

(2) Pvu I and Pst I

(3) Hind III and Sma I

(4) Pvu II and BamH I

54) Read the given below statements:
(i) Thermostable DNA polymerase is isolated from a bacterium, *Thermus aquatics*.
(ii) PCR is used for amplification of gene of intererst.
(iii) Multiple copies of DNA is synthesised in-vivo using PCR.
Find out correct statements:

(1) i and iii

(2) i and ii

(3) i, ii and iii

(4) ii and iii

55) Transgenic models exist for which of the following human diseases :
(a) Cancer
(b) Cystic fibrosis
(c) Rheumatoid arthritis
(d) Alzheimer's disease

(1) only a, b and c

(2) a, b, c, d

(3) only b, c, d

(4) only a and b

56) Recombinant DNA technology involves several steps in specific sequence. What is the correct sequence for r-DNA technology ?
a → Down stream processing.
b → Cutting of DNA at specific location.
c → Obtaining the foreign gene product.
d → Isolation of the genetic material.
e → Insertion of r-DNA into the host cell.

(1) b → d → e → a → c

(2) d → b → e → c → a

(3) d → e → d → a → c

(4) e → d → a → c → b

57) Match the column A from column B

	Column-A		**Column-B**
(I)	ELISA	(a)	Nematode resistance plant
(II)	RNA-i	(b)	Cotton bollworm
(III)	Cry IAC	(c)	Molecular diagnosis
(IV)	α-1 antitrypsin	(d)	To treat emphysema

(1) I → c, II → a, III → b, IV → d

(2) I → c, II → b, III → d, IV → a

(3) I → a, II → b, III → c, IV → d

(4) I → d, II → c, III → a, IV → b

58) **Statement-I:** If any protein encoding gene is expressed in heterologous host, is called recombinant protein.
Statement-II: Cells in continuous culture system remain in physiologically most active lag phase.

(1) Both statement-I and statement-II are correct.

(2) Both statement-I and statement-II are incorrect.

(3) Statement-I is correct but statement-II in incorrect.

(4) Statement-I is incorrect but statement-II is correct.

59) Two statements are given below :
Statement-I : Retro viruses in animals have the ability to transform normal cell into cancerous cells.
Statement-II : *Agrobacterium tumifaciens*, a pathogen of several dicot plants is able to deliver a piece of DNA known as 'T-DNA' into plants.

(1) Both statements I and II are true

(2) Both statements I and II are false

(3) Statement-I is true and statement-II is false

(4) Statement-I is false and statement-II is true

60) How many of the following enzymes are not used for isolation of DNA from *E.coli*?
(A) Cellulase (B) Lysozyme
(C) Chitinase (D) Taq DNA polymerase
(E) EcoRI (F) DNA ligase
(G) Protease (H) Ribonulease

(1) Four

(2) Three

(3) Six

(4) Five

61) Read the given below statements:
(i) Biolistics method of gene transfer is suitable for animal cell.
(ii) Microinjection is a method of gene transfer into the nucleus of an animal cell.
(iii) Gene gun method is suitable for plant cell
(iv) Microparticles for gold or tungsten are used in biolistics method of gene transfer.
(v) In order to force bacteria to take up the plasmid the bacterial cell must first be made 'competent' to take up DNA.
How many above statement are correct?

(1) Two
(2) Three
(3) Four
(4) Five

62) The blue and white selectable markers have been developed which differentiate recombinant colonies from non-recombinant colonies on the basis of their ability to produce colour in the presence of a chromogenic substrate.
Given below ae two statements about this method:
Statement-I: The colonies do not produce any colour these are identified as recombinant colonies.
Statement-II: The blue coloured colonies are identified as recombinant colonies.
In the light of above statement, choose the most appropriate answer from the options given below:-

(1) Both statement-I & statement-II are correct
(2) Both statement-I & statement-II are incorrect
(3) statement-I is correct but statement-II is incorrect
(4) statement-I is incorrect but statement II is correct

63) Which of the following genetically engineered organisms was used by Eli Lilly to prepare human insulin ?

(1) Bacterium
(2) Yeast
(3) Virus
(4) Phage

64) A human protein α-1-antitrypsin is used to treat ?

(1) Phenylketonuria
(2) Haemophilia
(3) Colorblindness
(4) Emphysema

65) The method of producing thousands of plants through tissue culture is called..... .

(1) Somaclones
(2) Micropropagation
(3) Explant

(4) Macropropagation

66) Cry IAc and Cry IAb produce toxins that control respectively :-

(1) Cotton boll worms and corn borer
(2) Corn borer and cotton boll worms
(3) Tobacco bud worms and nematodes
(4) Nematodes and tobacco bud worms

67) The use of bioresources by multinational companies and other organization without proper authorisation from countries and people concered with compensatory payment it is known as :-

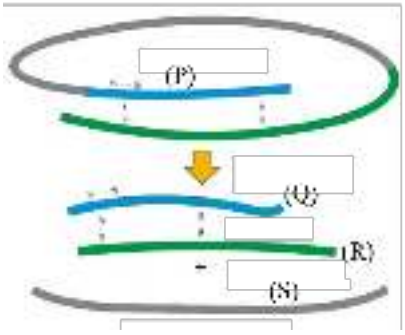
(1) Biopiracy
(2) Biopatent
(3) Biowar
(4) Bioremediation

68) The two polypeptides of human insulin are linked together by :-

(1) Hydrogen bonds
(2) Phosphodiester bond
(3) Covalent bond
(4) Disulphide bridges

69) **Assertion :** A novel strategy was adopted to prevent infestation of Nematodes in tobacco was based on the process of RNA interference (RNAi)
Reason : RNAi takes place in all prokaryotic organisms as a method of cellular defense.

(1) Both **(A)** and **(R)** are true and **(R)** is the correct explanation of **(A)**.
(2) **(A)** is true but **(R)** is false.
(3) Both **(A)** and **(R)** are true but **(R)** is NOT the correct explanation of **(A)**.
(4) **(A)** is false but **(R)** is true.



70) Read the following statements about the given diagram.
(i) P-Proinsulin secreted by pancreas, S-not formed when insulin is produced by recombinant DNA technology.
(ii) Q-Peptide chain is not present in the mature insulin.
(iii) Diagram represent the muturation of pro-insulin into insulin.
(iv) Insulin is made up of 51 amino acids.

How many statements are correct ?

(1) 1
(2) 2
(3) 3
(4) 4

71) Which of the following statement is incorrect regarding genetically engineered insulin ?

(1) The main challenge for production of insulin using rDNA techniques was getting insulin assembled into a mature form.
(2) Eli Lilly prepared two DNA sequences corresponding to A and B chains.
(3) Eli Lilly produced chain A and B separately.
(4) C-peptide was removed after creating disulfide bond.

72)

Which of the following shows correct chronological order of the events occuring during tissue culture ?

(1) Callus → Cell division → Explant → Growth regulators → Aquire meristamatic property
(2) Explant → Callus → Cell division → Growth regulators → Cell aquire meristamatic property
(3) Explant → Cell division → Callus → Growth regulators → Cell aquire meristamatic property
(4) Callus → Explant → Cell division → Growth regulators → Cell aquire meristamatic property

73) Match column I with column II and select the correct option from the given codes.

	Column I		**Column II**
(A)	Biopiracy	(i)	Effort to fix the non-functional gene
(B)	Biopatent	(ii)	Gene silencing
(C)	Gene therepy	(iii)	Illegal removal of biological materials
(D)	RNAi	(iv)	Right granted for biological entities

(1) A → iv, B → ii, c → i, D → iii
(2) A → ii, B → iv, c → i, D → iii
(3) A → iii, B → iv, c → i, D → ii
(4) A → iii, B → iv, c → ii, D → i

74) Match the following columns :-

Column-I		**Column-II**	
A	Totipotency	I	Genetically identical to original plant
B	Micro-propagation	II	Free from virus

C	Somaclones	III	Producing thousands of plants
D	Meristem	IV	Capacity to generate a whole plant from explant

(1) A - IV, B - I,C - III, D - II

(2) A - III, B - IV, C - II, D - I

(3) A - IV, B - III,C - I, D - II

(4) A - IV, B - I,C - II, D - III

75) **Statement - I :** Gene therapy aims to treat disease by correcting defective genes.
Statement - II : It involves insertion of a functional gene into patient's cells.

(1) Both Statement I and Statement II are incorrect.

(2) Statement I is correct but Statement II is incorrect.

(3) Statement I is incorrect but Statement II is correct.

(4) Both Statement I and Statement II are correct.

76) Given below are two statements :-
Statement-I : Genetic modification has reduced reliance on chemical pesticides.
Statement-II : The proteins encoded by the genes cryIAc and cryIIAb control corn bores.

(1) Both statement-I and statement-II are true.

(2) Both statement-I and statement-II are false.

(3) Statement-I is true but statement-II is false.

(4) Statement-I is false but statement-II is true.

77) How many of the following statement are correct ?
(A) Genetic modifications in plants may enhance nutritional value of food.
(B) Recombinant DNA technology can serve the purpose of early diagnosis of a disease.
(C) In RNAi (RNA interference) the source of complementary RNA could be transposons.
(D) Human protein α-1 antitrypsin is used to treat cystic fibrosis.

(1) One

(2) Two

(3) Three

(4) Four

78) Read the statements carefully.
(I) Basmati rice is distinct for its unique aroma and flavor and 27 documented varieties of basmati are grown in India.
(II) The Indian government has set up organisations such as GEAC.
(III) Probe is a single stranded DNA or RNA tagged with a radioactive molecule.
How many above given statement is/are correct :-

(1) Only I

(2) Only II

(3) Only I & II

(4) I, II and III

79) Which of the following statements are correct ?
A. Genetic modification has made crops intolerant to abiotic stress.
B. A nematode *Meloidegyne incognita* infects the leaves of tobacco plant.
C. Gene therapy is a collection of methods that allows correction of a gene defect.
D. Today transgenic models exist for many human disease such as cancer.
E. Toxicity testing in transgenic animals will allow us to obtain results in less time.
Choose the most appropriate answer from the options given below.

(1) B, C and D only

(2) C, D and E only

(3) A, B and E only

(4) A, B and D only

80) Match the column-I, column-II and column-III :

	Column-I		**Column-II**		**Column-III**
A	*Bacillus thuringiensis*	P	Molecular diagnosis	I	Eli Lilly
B	RNA interference	Q	Recombinant insulin	II	Bt cotton
C	*E.coli*	R	mRNA silencing	III	HIV test
D	PCR	S	Insecticidal protein	IV	Nematode resistant tobacco

Choose the correct answer from the options given below :

(1) A-S-II, B-R-IV, C-Q-I, D-P-III

(2) A-Q-I, B-P-IV, C-S-III, D-R-II

(3) A-Q-II, B-P-III, C-S-I, D-R-IV

(4) A-R-I, B-P-III, C-S-II, D-Q-IV

81) Given below are some steps of formation of insulin by genetic engineering :-
A. Chain A and B extracted separately.
B. Gene of A and B transfer in plasmids of E.coli to produce insulin chains.
C. Chain A and B combined by creating disulfide bonds.
D. Chains A and B were produced separately.
E. Prepared by two DNA sequences corresponding to A and B chains.
Choose the correct sequence of stages/steps from the options given below :-

(1) C-E-D-A-B

(2) E-B-D-A-C

(3) B-D-E-A-C

(4) E-C-A-D-B

82) Biological process is carried out in which type of water treatment :

(1) Primary
(2) Secondary
(3) Tertiary
(4) None of above

83) Penicillin is obtained from :-

(1) Yeast
(2) Bacteria
(3) Fungi
(4) Algae

84) The bottle juices are clarified by the use of :-

(1) Lipase
(2) Proteases
(3) Pectinases
(4) Both 2 and 3

85) Which of the following is also called brewer's yeast?

(1) *Penicillum notatum*
(2) *Acetobactoer aceti*
(3) *Saccharomyces cerevisiae*
(4) *Trichoderma*

86) Which of the following product is not obtained from fungus ?

(1) Cyclosporin A
(2) Citric acid
(3) Penicillin
(4) Butyric acid

87) **Assertion (A) :** Statins produce by the yeast have been commercialised as blood-cholesterol lowering agents.
Reason (R) : Statins competitively inhibit the enzyme responsible for synthesis of cholesterol.
Select the correct option :-

(1) Both (A) and (R) are correct but (R) is not the correct explanation of (A).
(2) (A) is correct but (R) is incorrect.
(3) Both (A) and (R) are correct but (R) is the correct explanation of (A).
(4) Both (A) and (R) are incorrect.

88) Match the following :-

	Column-I		**Column-II**

(a)	Citric acid	(i)	*Trichoderma polysporum*
(b)	Butyric acid	(ii)	*Aspergillus niger*
(c)	Cyclosporin-A	(iii)	*Monascus purpureus*
(d)	Statins	(iv)	*Clostridium butylium*

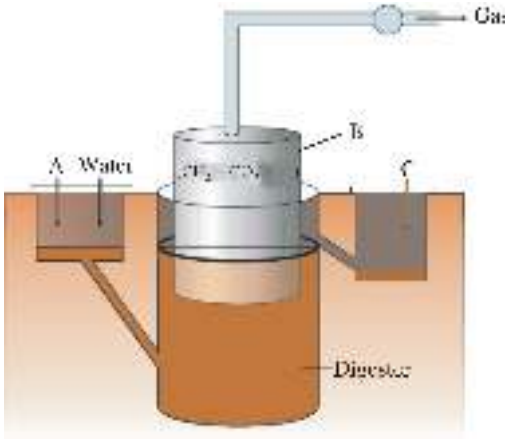
(1) a-iv, b-ii, c-i, d-iii

(2) a-ii, b-iv, c-i, d-iii

(3) a-ii, b-iv, c-iii, d-i

(4) a-iii, b-iv, c-i, d-ii

89) Identify A, B and C in the given diagram of typical biogas plant.



(1) A - Gas holder, B - Dung, C - Sludge

(2) A - Sludge, B - Dung, C - Gas holder

(3) A - Dung, B - Gas holder, C - Sludge

(4) A - Dung, B - Sludge, C - Gas holder

90) Read the following statements :-
(a) Depending on the type of the raw material used for fermentation and the type of processing different types of alcoholic drinks are obtained.
(b) Penicillin is obtained from bacteria *Penicillium notatum.*
(c) Antibiotics are against life with reference to human being
(d) Diphtheria can be cured by antibiotic.
(e) Microbes are used for commercial and industrial production of enzymes.
Choose the option which have all the correct statements only.

(1) a, b, c

(2) c, d, e

(3) b, d, e

(4) a, d, e

ANSWER KEYS

PHYSICS

Q.	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20
A.	1	4	2	2	2	3	4	2	2	1	2	4	3	2	3	2	4	4	4	1
Q.	21	22	23	24	25	26	27	28	29	30	31	32	33	34	35	36	37	38	39	40
A.	1	2	3	1	2	2	2	1	1	2	1	4	4	1	4	4	3	2	1	2
Q.	41	42	43	44	45															
A.	1	2	3	4	3															

CHEMISTRY

Q.	46	47	48	49	50	51	52	53	54	55	56	57	58	59	60	61	62	63	64	65
A.	2	3	1	1	4	4	3	2	2	3	2	2	2	2	2	4	4	4	3	4
Q.	66	67	68	69	70	71	72	73	74	75	76	77	78	79	80	81	82	83	84	85
A.	2	2	2	3	1	4	2	4	3	2	2	4	4	3	3	4	4	2	4	2
Q.	86	87	88	89	90															
A.	1	4	3	4	2															

BIOLOGY

Q.	91	92	93	94	95	96	97	98	99	100	101	102	103	104	105	106	107	108	109	110
A.	3	2	2	1	4	4	1	2	4	4	4	3	2	3	3	4	1	2	3	3
Q.	111	112	113	114	115	116	117	118	119	120	121	122	123	124	125	126	127	128	129	130
A.	2	3	1	3	4	1	3	2	2	3	1	3	3	4	2	4	2	4	2	2
Q.	131	132	133	134	135	136	137	138	139	140	141	142	143	144	145	146	147	148	149	150
A.	1	3	4	3	2	3	2	3	4	4	1	4	2	2	2	2	1	3	1	4
Q.	151	152	153	154	155	156	157	158	159	160	161	162	163	164	165	166	167	168	169	170
A.	3	3	1	4	2	1	1	4	2	3	4	3	3	3	4	3	3	4	2	1
Q.	171	172	173	174	175	176	177	178	179	180										
A.	2	2	3	4	3	4	3	2	3	4										

SOLUTIONS

PHYSICS

1) $\tau = MB \sin\theta \Rightarrow \tau = (mL) B \sin\theta$

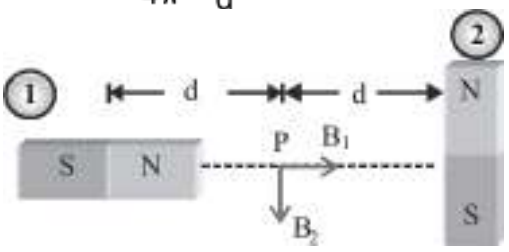
$\Rightarrow \quad 25 \times 10^{-6} = (m \times 5 \times 10^{-2}) \times 5 \times 10^{-2} \times \sin 30°$

$\Rightarrow \quad m = 2 \times 10^{-2}$ A – m

2) At point P, net magnetic field $B_{net} = \sqrt{B_1^2 + B_2^2}$

where, $B_1 = \frac{\mu_0}{4\pi}.\frac{2M}{d^3}$ and $B_2 = \frac{\mu_0}{4\pi}.\frac{M}{d^3}$

$\Rightarrow B_{net} = \frac{\mu_0}{4\pi}.\frac{\sqrt{5}M}{d^3}$



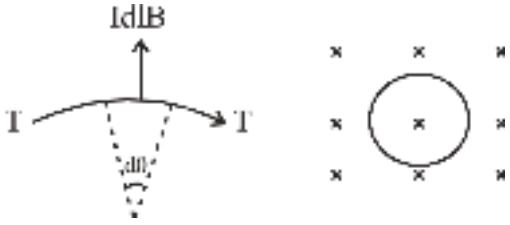
3) If radius of circle is r, then $2\pi r = L \Rightarrow r = \frac{L}{2\pi}$

Area = $\pi r^2 = \frac{\pi L^2}{4\pi^2} = \frac{L^2}{4\pi}$

magnetic moment = $IA = \frac{IL^2}{4\pi}$

4) $\vec{E} = -(\vec{V} \times \vec{B})$

5) $\vec{B} = \frac{\mu_0 i}{2\pi}\left[\left(\frac{1}{1} + \frac{1}{4} + \frac{1}{16} + ...\right)\hat{k} - \left(\frac{1}{2} + \frac{1}{8} + ...\right)\hat{k}\right] = 1.33 \times 10^{-7}\hat{k}$



6)

$2T\sin\left(\frac{d\theta}{2}\right) = IdlB \Rightarrow 2T\frac{dl}{2} = IRdlB$

$\Rightarrow T = BIR = \frac{BIL}{2\pi}$

7) $r = \frac{\sqrt{2qVm}}{Bq}$ or $r \propto \sqrt{\frac{m}{q}}\frac{r_p}{r_\alpha} = \sqrt{\frac{m_p}{m_\alpha}}\sqrt{\frac{q_\alpha}{q_p}} = \sqrt{\frac{1}{4}}\sqrt{\frac{2}{1}} = \frac{1}{\sqrt{2}}$.

8) $B = 10\ T = \frac{\mu_0 I}{2\pi R^2}.\frac{3R}{4}$

$\Rightarrow \frac{\mu_0 I}{2\pi R} = \frac{40}{3}\ T$

$B^1 = \frac{\mu_0 I}{2\pi(5R)} = \frac{1}{5} \times \frac{40}{3} = \frac{8}{3} T$

9) B_1 = due to semi infinite wire = $\frac{\mu_0 I}{4\pi R}(-\hat{k})$

B_2 = due to semi infinite wire = $\frac{\mu_0 I}{4\pi R}(-\hat{k})$

B_3 = due to semi-circle = $\frac{\mu_0 I}{4R}(-\hat{i})$

$B_{Net} = \sqrt{(B_1 + B_2)^2 + B_3^2} = \frac{\mu_0 I}{4\pi a} \times \sqrt{4 + \pi^2}$

10) $B_x = \frac{\mu_0 I a^2}{2(x^2 + a^2)^{\frac{3}{2}}}$

$B_{center} = \frac{\mu_0 I}{2a}$

$B_x = \frac{1}{2\sqrt{2}}\frac{\mu_0 I}{2} = \frac{\mu_0 I}{2(x^2+1)^{3/2}}$

$\frac{1}{8} = \frac{1}{(x^2+1)^3}$

$x^2 = 1$

$x = \pm 1$

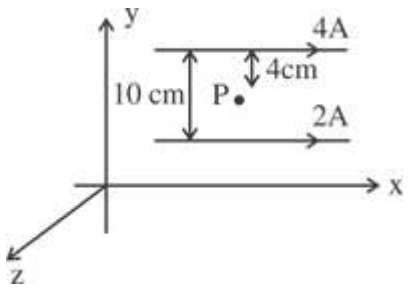
11) $\mu_r < 1$

$\varepsilon_r > 1$

12) $\frac{B_1}{B_2} = \frac{1}{3} = \frac{\mu_0 I_1 \times 2R_2}{2R_1 \times \mu_0 I_2}$

$\frac{1}{3} = \frac{I_1}{I_2} \times \frac{R_2}{R_1}$

$\frac{I_1}{I_2} = \frac{1}{3} \times \frac{R_1}{R_2} = \frac{1}{3} \times \frac{1}{2} = \frac{1}{6}$



13) $B_{net} = B_1 - B_2 = \frac{\mu_0 \times 4}{2\pi[.04]} - \frac{\mu_0 \times 2}{2\pi[.06]}$

$\vec{B}_{net} = \frac{\mu_0}{2\pi}\left[\frac{200}{3}\right](-\hat{k})$

$\vec{F} = q\left[\vec{v} \times \vec{B}\right]$

$= [3\pi]\left[\left(2\hat{i} + 3\hat{j}\right) \times \left(\frac{\mu_0}{2\pi}\right)\left(\frac{200}{3}\right)(-\hat{k})\right]$

$= 3\pi \times \frac{\mu_0}{2\pi}\left(\frac{200}{3}\right)\left[2 \times \hat{j} - 3\left(\hat{i}\right)\right]$

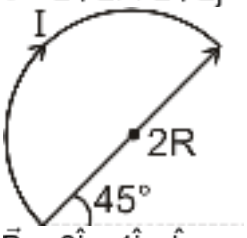
$= \left(4\pi \times 10^{-7}\right)(100)\left(-3\hat{i} + 2\hat{j}\right)$

$= 4\pi \times 10^{-5} \times \left[-3\hat{i} + 2\hat{j}\right]$

14) $\vec{\ell} = 2R\cos 45^\circ \hat{i} + 2R\sin 45^\circ \hat{j}$

$\vec{\ell} = \sqrt{2}R\hat{i} + \sqrt{2}R\hat{j}$

$\vec{\ell} = 2\sqrt{2}\hat{i} + 2\sqrt{2}\hat{j}$



$\vec{B} = 3\hat{i} + 4\hat{j} - \hat{k}$

$\vec{F} = I\left(\vec{\ell} \times \vec{B}\right)$

$\Rightarrow \sqrt{2}\left(2\sqrt{2}\hat{i} + 2\sqrt{2}\hat{j}\right) \times \left(3\hat{i} + 4\hat{j} - \hat{k}\right)$

$\vec{F} \Rightarrow -4\hat{i} + 4\hat{j} + 4\hat{k}$

15)

from ACL, $\oint \vec{B}.d\vec{\ell} = \mu_0 i_{enc.}$

only a quarter of thick wire is inscribed by square

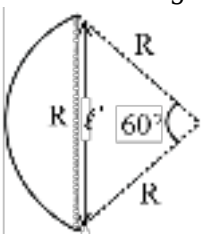
$\Rightarrow i_{enc} = i/4$

$\Rightarrow \oint \vec{B} \cdot d\vec{\ell} = \frac{\mu_0 i}{4}$

16) $\frac{x_2}{x_1} = \frac{T_1}{T_2} \Rightarrow T_2 = 200$ K

17)

For straight wire : M = m × ℓ
After bending the wire :



M' = m × ℓ' = m × R

where, $R = \frac{\ell}{(\pi/3)}$

So, $M' = \frac{3m\ell}{\pi} = \frac{3M}{\pi}$

18) $P = \frac{V^2}{R}$

$50 = \frac{100 \times 100}{R}$

R = 200 Ω

Max current that can be passed through bulb without fusing it,

V = iR

100 = i × 200

i_{max} = 0.5 A

For AC, $I_{max} = \frac{V}{\sqrt{R^2 + (2\pi fL)^2}}$

$0.5 = \frac{200}{\sqrt{4 \times 10^4 + (2\pi \times 50 \times L)^2}}$

$4 \times 10^4 + (100\ \pi L)^2 = 16 \times 10^4$

$(100\ \pi L)^2 = 16 \times 10^4 - 4 \times 10^4$

$(100\ \pi L)^2 = 12 \times 10^4$

$100\ \pi L = 2\sqrt{3} \times 100$

$L = \frac{2\sqrt{3}}{\pi} = \frac{1.732 \times 2}{3.14} = 1.1H$

19)

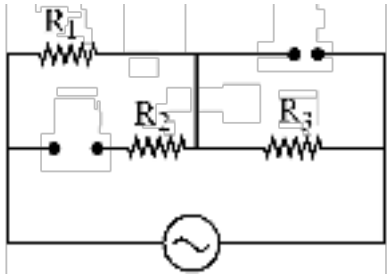
Bulb is purely resistive circuit, so equally bright.

20)

Frequencies → Very high

ℓ L → Open circuit, and

C → Wire



$\therefore R_{eq} = R_1 + R_3$

21)

$$e = \left(\frac{e_1 \sin \omega t}{\sqrt{e_1^2 + e_2^2}} + \frac{e_2 \cos \omega t}{\sqrt{e_1^2 + e_2^2}} \right) \sqrt{e_1^2 + e_2^2}$$

Let, $\cos\phi = \frac{e_1}{\sqrt{e_1^2 + e_2^2}}$ $\sin\phi = \frac{e_2}{\sqrt{e_1^2 + e_2^2}}$

then, $e = \sqrt{e_1^2 + e_2^2} \sin(\omega t + \phi)$

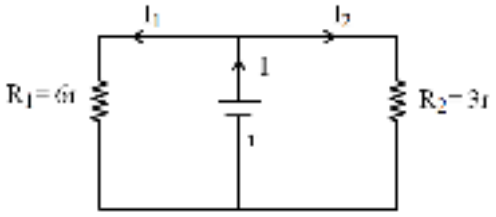
$\therefore$ $e_{rms} = \frac{\sqrt{e_1^2 + e_2^2}}{\sqrt{2}} = \sqrt{\frac{e_1^2 + e_2^2}{2}}$

22) Frequency of power = 2 times the frequency of voltage
$= 2 \times 50 = 100\text{Hz}$

23)

$X_L = X_C$
$\therefore$ Resonant circuit, so, power factor = 1

24) $\because X_L << X_C$
So, bulb A will glow brighter.

25)


$$R_T = r + \frac{3r \times 6r}{3r \times 6r} = 3r$$

$$I_T = \frac{BVL}{3r} \text{.....(1)}$$

$$I_1 = \frac{R_2}{R_1 + R_2} \times I_T = \frac{3r}{9r} \left(\frac{BVL}{3r} \right)$$

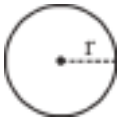
$$I_1 = \frac{1}{3} \left(\frac{BVL}{3r} \right)$$

$I_2 = \left(\frac{R_1}{R_1 + R_2}\right) I = \frac{6r}{9r}\left(\frac{BVL}{3r}\right)$

$I_2 = \frac{2}{3}\left(\frac{BVL}{3r}\right)$

26) $\phi = B.A.$

$\phi = B(Ir^2)$



emf (e) = $\frac{d\phi}{dt}$

$e = \frac{d}{dt}(B\pi r^2)$

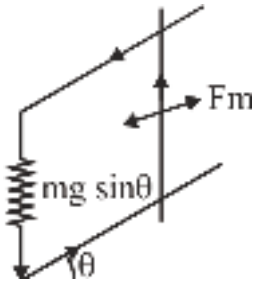
$e = 2B\pi r \frac{dr}{dt}$

From t = 0 to t = 1s, $\frac{dr}{dt} = 0$

From t = 1s to t = 2s, $\frac{dr}{dt}$ = constant but r increases linearly.

27) Terminal velocity is attained when magnetic force is equal to mg sin θ

 $F_m = mg \sin\theta$



$F_m = mg \sin\theta$

or $\left(\frac{e}{R}\right)\ell B = mg \sin\theta$

or $\frac{(B\nu_T\ell)}{R} \times \ell B = mg \sin\theta$

$\therefore \nu_T = \frac{mgR\sin\theta}{B^2\ell^2}$

28)

Step 1: Calculate the induced emf:

$\varepsilon = BvL = 4T \times 2\ m/s \times 1.0\ m = 8\ V$

Step 2: Calculate the charge stored in the capacitor:

$Q = CV = 10 \times 10^{-6}\ F \times 8V = 80 \times 10^{-6}\ C = 80\ \mu C$

Step 3: Determine the polarity: Positive charges accumulate at the end of the rod moving towards terminal P (and therefore terminal A), while negative charges accumulate on terminal B.

29)

As magnet comes close to coil flux through coil increases so an emf is induced. When magnet goes out from the coil flux through coil decreases so again emf is induced but in opposite with opposite polarity. This process repeats.

30)

Electrostatic field lines never form a closed loop.

31) According to Lenz's law, the emf induced in coil is such that it resists the motion of the magnet. As emf in each turn opposes the motion of the magnet, when the number of turns is increased the force opposing the motion also increases. Thus, pushing a magnet into a coil with more number of loops is more difficult.

32) $M = \frac{\phi_2}{I_1} = \frac{B_1 A_2}{I_1}$

$= \frac{\left(\frac{\mu_0 I_1}{2R_1}\right) \pi R_2^2}{I_1}$

$M \propto \frac{R_2^2}{R_1}$

33) Effective length of the loop = 2R
perpendicular component of velocity to effective length vsinθ
e = $B(\vec{\ell} \times \vec{v})$
e = B2R v sin θ

34)

$Q = \int i\, dt$ = Area = $\frac{1}{2}(4)(0.1)$ = 0.2 C

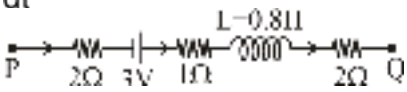
$Q = \frac{\Delta\phi}{R}$

$\Delta\phi = QR$ = (0.2) (10) = 2 wb

35)

I = 3A

$\frac{dI}{dt} = 2As^{-1}$



$v_P - 3 \times 2 + 3 - 3 \times 1 - 0.8 \times 2 - 3 \times 2 = v_Q$

$v_P - v_Q = 13.6$ V

36)

Gauss Theorem in electrostatics

$$\oint \vec{E} \cdot \vec{ds} = \frac{Q}{\varepsilon_0}$$

Gauss Theorem in magnetism

$$\oint \vec{B} \cdot \vec{ds} = 0$$

Faraday Law

$$\oint \vec{E} \cdot \vec{d\ell} = \frac{-d\phi}{dt}$$

Ampere-Maxwell's Law

$$\oint \vec{B} \cdot \vec{d\ell} = \mu_0 i_c + \mu_0 \varepsilon_0 \frac{d(\phi_E)}{dt}$$

37) $C = \frac{\varepsilon_0 KA}{d} = \frac{(8.85 \times 10^{-12}) \times 10 \times 1}{10^{-3}}$

$i = \frac{d}{dt}(CV) = C\frac{dV}{dt} = 8.85 \times 10^{-8} \times 25$

$= 2.2 \times 10^{-6} A$

38)

EMW propagating along + Z direction so $\vec{E} \times \vec{B}$ would be along + Z direction and $\vec{E} \perp \vec{B}$, option (2) satisfy both conditions.

39)

- The energy Of electromagnetic radiation determines its wavelength and spectral region.
- Approximate energy ranges for different regions:
- Visible light: 1.65 – 3.1 ev
- Ultraviolet 3 -124 ev
- X-rays: 124 ev – 100 kev
- Gamma rays (γ-rays): > 100 kev

Formula:

photon energy relation:

$$E = \frac{hc}{\lambda}$$

where h is Planck's constant, c is the speed of light and X is the wavelength.

Conclusion:

- 14.4 ev falls in the range of ultraviolet (UV) radiation.

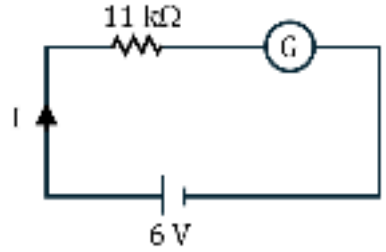
40) Excimer lasers used in LASIK operate at a wavelength of about 193 nm, which lies in the ultraviolet region of the electromagnetic. spectrum. This radiation allows precise ablation without excessive heating.

41) $|\vec{S}| = \frac{EB}{\mu_0} = \frac{E_0^2}{2C\mu_0}$

$= \frac{10^4}{2 \times 3 \times 10^8 \times 4\pi \times 10^{-7}} = 13.25\ Wm^{-2}$

42)

If no shunt

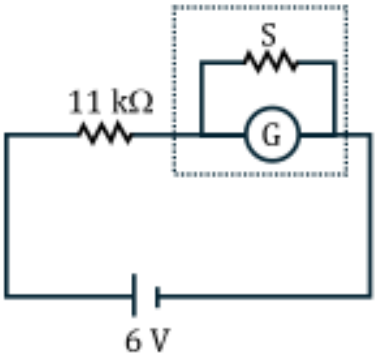


$I = 9 \times 60\ \mu A = 540 \times 10^{-6}\ A$

$$\frac{6}{11000+G} = 540 \times 10^{-6}$$

$$G = \frac{1000}{9}$$

If shunt is applied



From the standard relation from half deflection method.

$$G = \frac{RS}{R-S} \Rightarrow \frac{1000}{9} = \frac{11000\,S}{11000-S}$$

$S = 110\ \Omega$

43)

$$\frac{\theta}{i} = \frac{5}{10^{-3}} = 5000 \quad \frac{\theta}{V} = 20$$

$$\Rightarrow \frac{\theta/i}{\theta/V} = \frac{5000}{20} = 250$$

$$\Rightarrow \frac{V}{i} = R = 250\Omega$$

44) • Statement (A) is correct: A choke coil needs high inductance L to limit current and low resistance R to minimize heat loss.
• Statement (B) is correct: Choke coils have a power factor cos ϕ ≈ 0, so, they consume negligible power compared to resistive rheostats.
• Statement (C) is incorrect: Inductive reactance XL is directly proportional to frequency, not inversely.
• Statement (D) is incorrect: Capacitive reactance X_C is inversely proportional to frequency, not directly.
Calculation inductive Reactance:
$X_L = 2\pi f L \Rightarrow X_L \propto f$ and

capacitive Reactance:

$X_C = \frac{1}{2\pi fC} \Rightarrow X_C \alpha \frac{1}{f}$

Average power :

$P = V_{rms} I_{rms} \cos \phi$

45)

$p = vT$

• The time period of circular motion is:

$T = \frac{2\pi m}{qB}$

• Since $v = \frac{v}{\sqrt{2}}$ we substitute:

$p = \frac{v}{\sqrt{2}} . \frac{2\pi m}{qB} \Rightarrow v = \frac{pqB}{2\pi m}\sqrt{2}$

$r = \frac{mv}{\sqrt{2}qB}$

Substituting v:

$r = \frac{m}{\sqrt{2}qB} \times \frac{pqB}{2\pi m}\sqrt{2} \quad r = \frac{p}{2\pi}$

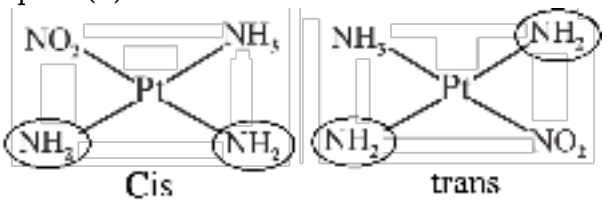
CHEMISTRY

46)

Theory information

47)

Option (1) are structural, option (2) are not isomers because the molecular formula is different option(3) are Cis and trans



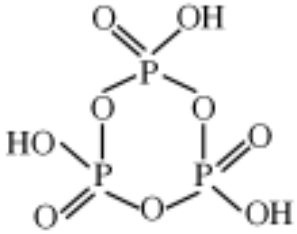
48)

Since copper is below hydrogen in SRP series, it can not reduce hydrogen and hence cannot liberate H_2 from acids

Eg. $Zn + dil.\ H_2SO_4 \rightarrow ZnSO_4 + H_2$

$Cu + dil.\ H_2SO_4 \rightarrow X$

49) $H_3P_3O_9$



3 double bonds and 12 single bonds

50) Due to small size and high IP of Be, Mg they do not give flame test.
Size of $Be^{+2} < Mg^{+2}$ hence hydration energy of $Be^{2+} > Mg^{2+}$.

51)

Stable $T\square I_3$ means $T\square^{+}$, I_3^{-}

52)

NCERT Data

53) **Question explanation:**
The question is asking about correctness of absorption wavelength order and ligand strength reasoning.
Concept: Ligand strength increases Δ_O; larger $\Delta_O \rightarrow$ absorption shifts to shorter wavelength (higher energy). Spectrochemical series: $H_2O < NH_3 <$ en.
Solution
Given order of absorption: $[Ni(H_2O)_6]^{2+} > [Ni(NH_3)_6]^{2+} > [Ni(en)_3]^{2+}$ implies H_2O absorbs at longest wavelength → weakest ligand, en strongest. The reason states $H_2O > NH_3 >$ en, which is incorrect. So (A) is correct, but (R) is wrong.
Final answer : option 2

54)

ppt. ⇒ $BaSO_4$, $PbSO_4$, AgI, AgBr, AgCl, $SrSO_4$, $CaCO_3$, $PbCrO_4$

55)

Octahedral Field (CN = 6)

$\overline{dx^2-y^2}$ $\overline{dz^2}$ eg. (High Energy)

$\overline{dxy}$ $\overline{dyz}$ $\overline{dzx}$ t_{2g} (Low Energy)

56) The major components in "Gun metal" are
Cu + Sn + Zn

57)

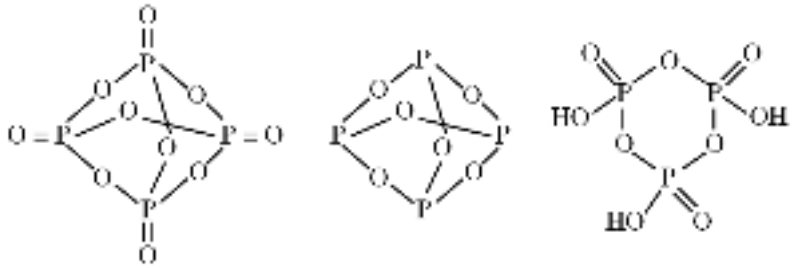
Since HCl itself is a reducing agent can react with $KMnO_4$ (O.A)

58)

In Thorium, last e^- enters in d subshell;
$_{90}Th = [Rn]\ 7s^2 6d^2$

59) $XeF_6 \xrightarrow{+H_2O} XeOF_4 \xrightarrow{+H_2O} \underset{(Y)}{XeO_2F_2} \xrightarrow{+H_2O} \underset{(X)}{XeO_3}$

60)



61) Acidic Strength :- HF < HCl < HBr < HI
HOF > HOCl > HOBr > HOI

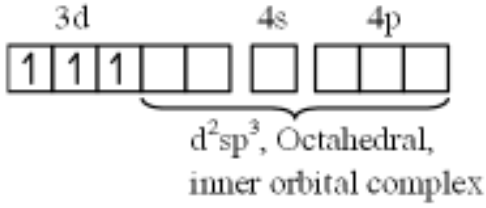
62)

$BaCl_2 + AgNO_3 \rightarrow AgCl$ (White ppt)
$BaCl_2 + ZnSO_4 \rightarrow BaSO_4$ (White ppt)
$ZnSO_4 + NaOH \rightarrow Zn(OH)_2$ (White ppt)

63)

$[CoBr_2(NH_3)_4]Cl \rightarrow [CoBr_2(NH_3)_4]^+ + Cl^- \Rightarrow$ 2 ions
↓ ↓
No AgBr ppt AgCl ppt

64) $[Cr^{+3}(NH_3)_6]Cl_3 + AgNO_3 \rightarrow AgCl\downarrow$
white
$Cr^{+3} = [Ar]4s^0\ 3d^3$

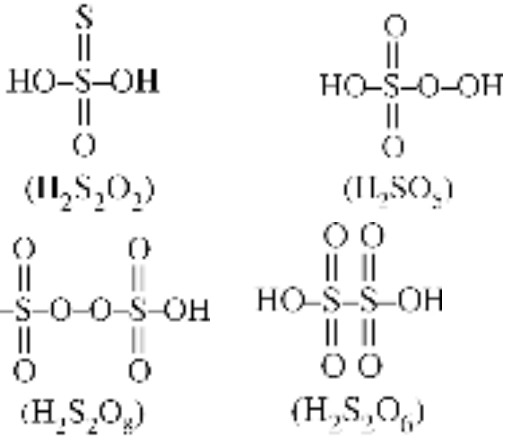


65) Only Bromide and Iodide gives yellow ppt. with $AgNO_3$ solution.

66)

$0 + x - 1 - 1 + x - 0 - 0 - (2 \times 2) = 0$
$\Rightarrow x = +3 \Rightarrow CN = 6$ (Co^{+3} here)

67) All lanthanoids do not show + 4 oxidation state.

68) 

69)

Due to steric crowding BCl_3, BBr, BI_3 does not form dimer.

70)  bond is weaker than  bond.

71)

$Cl^-/Br^-/F^- + I_2 \rightarrow X$ (No reaction)

72) NCERT, Pg. # 211 (OLD)

73)

NCERT, Pg. # 170, 171 (OLD-2019)

74)

$CaC_2 + H_2O \rightarrow Ca(OH)_2 + C_2H_2$
$Li_3N + H_2O \rightarrow Li(OH)_2 + NH_3$
$B_3N_3H_6 + H_2O \rightarrow B(OH)_3 + NH_3 + H_2$
$Al_4C_3 + H_2O \rightarrow Al(OH)_3 + CH_4$

75)

$ZnCl_2 + H_2S \rightarrow ZnS$ (Dirty White)
$CuSO_4 + NH_4OH \rightarrow [Cu(NH_3)_4]$ (Deep Blue)
$FeCl_3 + KSCN \rightarrow Fe(SCN)_3$ (Blood Red)
$Pb(NO_3)_2 + KI \rightarrow PbI_2$ (Yellow)

76) **Explaining the Question :-**Question is asking about what will be observed if we add with shaking, excess dil. NaOH solution $ZnCl_2$ solution.
Concept :-Colour of precipitate, Colour of solution, general reaction
Solution :- A white gelatinous precipitate forms and so white precipitate appears.Zn^{2+} (aq) + $2OH^-$ (aq) $\rightarrow Zn(OH)_2$Zinc hydroxide precipitates.The white precipitate **dissolves**, forming a **clear, colorless solution**.
Final Answer :- 2

77)

$M_2SO_4.M_2'(SO_4)_3.24H_2O$
$[M(H_2O)_6]^+ : [M'(H_2O)_6]^{+3} : SO_4^{-2}$
Li does not have vacant d & due to small size can't attach $6H_2O$ molecules.

78)

NO_2 dimerise at low temperature

79)

$SO_3 + K_2Cr_2O_7 \rightarrow X$ (No reaction)
Cl_2 disproportionation in basic medium concept.

80) Cu^{2+} is more stable because more –Ve $\Delta_{hyd}H°$ of Cu^{+2}, compensates second ionisation enthalpy of Cu.

81)

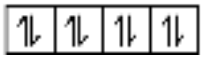
Concept of back bond

82) No free S^{2-} ion available in acidic medium of H_2S.

83)

Theory information

84)

$[Ni(CO)_4] \Rightarrow$  $\Rightarrow$ 0 unpaired e^-

$[Ni(H_2O)_6]Cl_2 \Rightarrow Ni^{+2} \Rightarrow$  $\Rightarrow$ 2 unpaired e^- (WFL)

$Na_2[Ni(CN)_4] \Rightarrow Ni^{+2} \Rightarrow$  $\Rightarrow$ 0 unpaired e^- (SFL)

$[PdCl_2(PPh_3)_2] \Rightarrow Pd^{+2} \Rightarrow$  $\Rightarrow$ 0 unpaired e^- (SFL)

85)

NCERT, Coordination compounds.

86)

$Ga^+ < Ga^{+3}$
$Bi^{+3} > Sb^{+3}$
$Sn^{+4} > Sn^{+2}$
$Pb^{+2} > Pb^{+4}$

87)

NCERT, p-block (Class-XII)

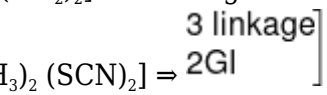
88) $CrO_4^{-2} \xrightarrow{H^+} Cr_2O_7^{-2}$

89)

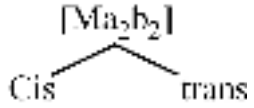
Concept

90)

$[Pt(en)(NO_2)_2] \Rightarrow$ 3 linkage isomers

$[Pd(NH_3)_2(SCN)_2] \Rightarrow$ 

6 isomers



Total isomers = 9

BIOLOGY

91)

NCERT-XII, Pg. # 57

92)

NCERT-XII, Pg. # 58

93)

NCERT-XII, Pg. 62

94)

The term phenocopy refers to an individual whose phenotype has been altered by environment to a state that resembles a phenotype associated with another genotype. Phenocopy is not inheritable.

95)

NCERT XIIth Pg.# 59

96)

NCERT XIIth Pg.# 67

97)

NCERT XIIth Pg.# 67

98)

NCERT XIIth Pg.# 73

99)

NCERT XIIth Pg.# 57, 59, 63

100)

NCERT XIIth Pg.# 55

101)

NCERT XIIth Pg.# 61, 62

102)

NCERT XIIth Pg.# 73

103)

NCERT XIIth Pg.# 62, 63

104) NCERT, Pg. # 59

105) NCERT XIIth Pg.# 67

106) NCERT XIIth Pg.# 61

107) NCERT XIIth Pg.# 62

108)

NCERT XIIth Pg.# 91, 94

109)

NCERT XIIth Pg.# 79

110)

NCERT XIIth Pg.# 92

111)

NCERT-XII, Pg. # 83

112)

NCERT XIIth Pg.# 80

113)

NCERT XIIth Pg.# 80

114)

NCERT XIIth Pg.# 89

115)

NCERT XIIth Pg.# 81, 82

116)

NCERT XIIth Pg.# 81

117)

NCERT XIIth Pg.# 97

118)

NCERT XIIth Pg.# 84

119)

NCERT XIIth Pg.# 83

120)

NCERT XIIth Pg.# 89

121)

NCERT XIIth Pg.# 91

122) NCERT XIIth Pg.# 97, 98

123) NCERT XIIth Pg.# 106

124)

NCERT XIIth Pg.# 90, 97, 101

125)

NCERT XIIth Pg.# 100, 101

126)

NCERT-XII, Pg. # 96, 100

127) NCERT XIIth Pg.# 99

128)

NCERT XIIth Pg.# 73

129)

NCERT XIIth Pg.# 62

130)

NCERT XIIth Pg.# 74

131)

NCERT XIIth Pg.# 72

132)

NCERT XIIth Pg.# 98

133)

NCERT XIIth Pg.# 87

134)

NCERT XIIth Pg.# 91

135)

NCERT XIIth Pg.# 80

136) NCERT-XII, Pg.#195

137)

NCERT-XII, Pg.#171

138)

NCERT-XII, Pg. # 169

139)

NCERT-XII, Pg. # 170

140)

NCERT-XII, Pg. # 169

141)

NCERT-XII, Pg. # 168

142) NCERT-XII, Pg. # 170

143) NCERT-XII, Pg.#169.

144) NCERT-XII, Pg. # 172

145)

NCERT-XII, Pg. # 183

146)

NCERT-XII, Pg. # 171

147) NCERT-XII, Pg. # 180,183,184

148) NCERT-XII, Pg. # 173

149) NCERT-XII, Pg. # 170

150) NCERT-XII, Pg. # 171
A,C,D,E,F are not used

151) NCERT-XII, Pg. # 170-171

152) NCERT-XII, Pg. # 170

153) NCERT-XII Pg # 182

154)

NCERT-XII, Pg. # 184

155)

NCERT-XII, Pg. # 178

156) NCERT-XII, Pg. # 180

157)

NCERT-XII, Pg. # 185

158)

NCERT-XII, Pg. # 182

159) NCERT-XII, Pg. # 180

160) NCERT-XII, Pg. # 184, 185

161) NCERT-XII, Pg. # 182(E), 193(H)

162)

NCERT-XI, Pg. # 178

163) NCERT-XII, Pg. # 185

164) NCERT-XII, Pg. # 178

165) NCERT-XII, Pg. # 182

166) NCERT-XII, Pg. # 179, 180 (E), 196, 197 (H)

167) NCERT-XII, Pg. # 179, 180, 181, 182

168)

NCERT-XII, Pg. # 183-185

169) NCERT-XII, Pg. # 179, 180, 182, 183, 184

170) NCERT-XII, Pg. # 179, 180, 182, 183

171) NCERT-XII, Pg. # 182

172)

NCERT-XII, Pg. # 156

173)

NCERT-XII, Pg. # 152

174)

NCERT-XII, Pg. # 153

175)

NCERT-XII, Pg. # 152

176)

NCERT-XII, Pg. # 153

177) NCERT-XII, Pg. # 153 (E), 168-169 (H)

178) NCERT-XII, Pg. # 153 (E), 168 (H)

179) NCERT-XII, Pg. # 156

180) NCERT-XII, Pg. # 152, 153 (E)/167, 168 (H)